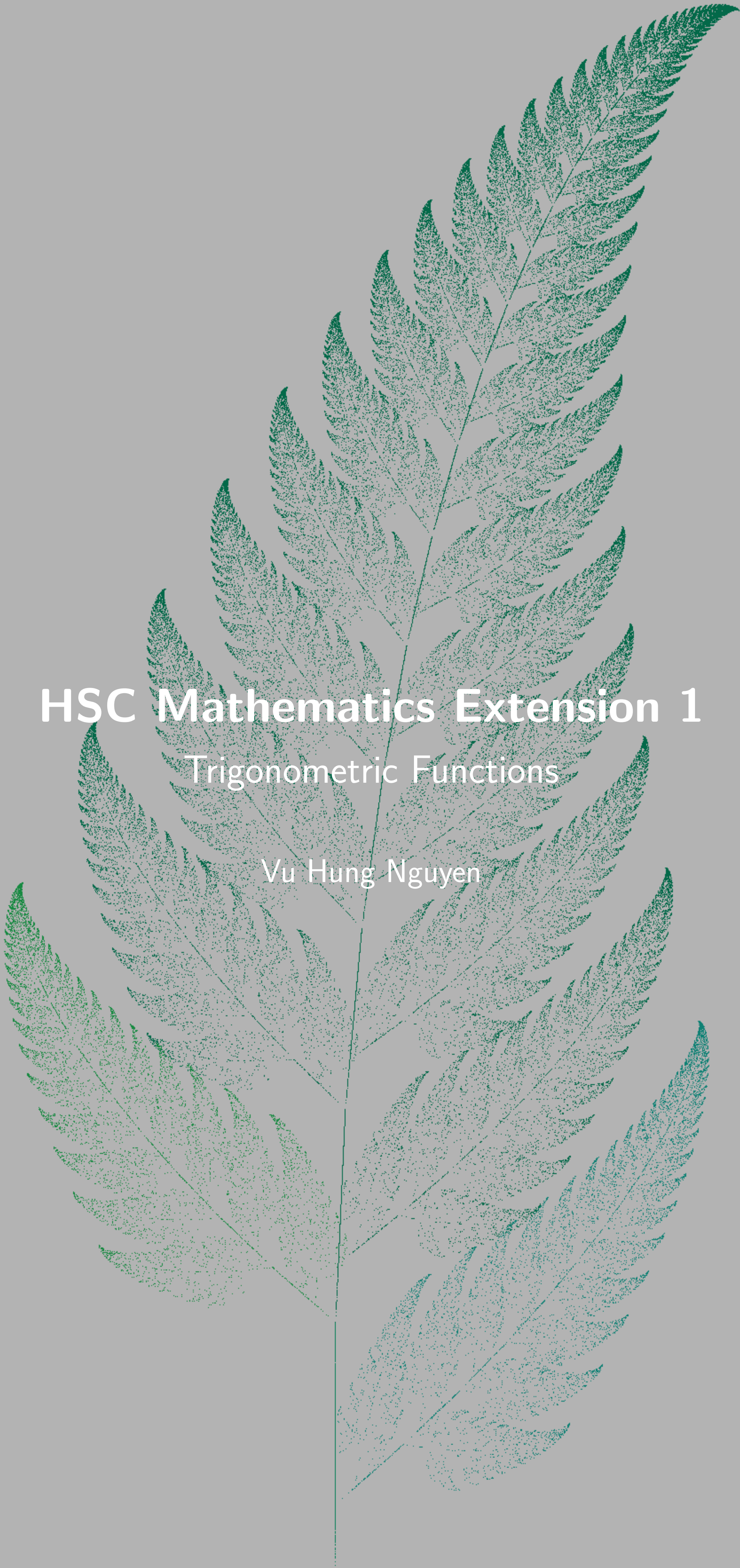


HSC Mathematics Extension 1

Trigonometric Functions

Vu Hung Nguyen

"Everything that has a beginning has an end."

The Matrix

Contents

1 Introduction

1.1 Project Overview

This booklet is a growing collection of trigonometry problems for NSW HSC Mathematics Extension 1 and Extension 2 students. It covers the full trigonometry strand across both syllabuses, including right-angled triangles, trigonometric functions of a general angle, trigonometric identities and equations, the sine and cosine rules, inverse trigonometric functions, compound and double-angle formulae, the t-formulae, and products-to-sums conversions. The problems range from straightforward skill-drill questions to multi-step challenge problems suitable for exam preparation and beyond.

1.2 Target Audience

This resource is written for students who want focused, exam-oriented practice with trigonometry in a clear, classroom-friendly format. Solutions are intentionally concise but each one highlights the key idea behind the answer. Students who work systematically through Parts 1 and 2 should gain both fluency and a deeper conceptual understanding of the trigonometry topics tested in Extension 1 and Extension 2.

1.3 How to Use This Booklet

- Read the core formulas and theory summaries in this introduction first.
- Attempt each problem before reading the solution.
- Pay attention to the *type* of question: is it an identity, an equation, a geometry application, or an inverse trigonometry question?
- Look for structure: symmetry, compound-angle patterns, Pythagorean identities, or opportunities to substitute $t = \tan(\theta/2)$.
- Use the remarks and takeaways to build pattern recognition for future HSC questions.

1.4 Extension 1 Knowledge Needed

Most problems in this booklet can be solved using standard HSC Mathematics Extension 1 and Extension 2 knowledge. The key background material is:

- **Right-angled trigonometry:** definitions of $\sin\theta$, $\cos\theta$, $\tan\theta$ as ratios of opposite, adjacent, and hypotenuse; exact values for $\pi/6$, $\pi/4$, $\pi/3$.
- **ASTC rule (quadrant signs):** All, Sine, Tangent, Cosine are positive in quadrants I, II, III, IV respectively.
- **Trigonometric functions of a general angle:** converting angles to the related acute angle and applying the correct sign.
- **Fundamental identity:** $\sin^2\theta + \cos^2\theta = 1$ and its derived forms $1 + \tan^2\theta = \sec^2\theta$ and $1 + \cot^2\theta = \csc^2\theta$.
- **Double and symmetry identities:** supplementary $(\pi - \theta)$, complementary $(\pi/2 - \theta)$, and reflex $(2\pi - \theta)$ angle reductions.

- **The sine rule and cosine rule:** applied to general (non-right-angled) triangles, including the ambiguous case.
- **Area formula:** Area $= \frac{1}{2}ab\sin C$.
- **Inverse trigonometric functions:** restricted domains for arcsin, arccos, arctan and graphical properties.
- **Compound-angle identities:** $\sin(\alpha \pm \beta)$, $\cos(\alpha \pm \beta)$, $\tan(\alpha \pm \beta)$.
- **Double-angle formulae:** $\sin 2\theta$, $\cos 2\theta$ (all three forms), $\tan 2\theta$.
- **The t-formulae:** expressing $\sin\theta$, $\cos\theta$, $\tan\theta$ in terms of $t = \tan(\theta/2)$.
- **Products-to-sums and sums-to-products:** converting between products and sums of trigonometric functions.

1.5 Special Angles

The exact values of the trigonometric functions for the special angles 0, $\pi/6$, $\pi/4$, $\pi/3$, and $\pi/2$ come from two standard right-angled triangles and the unit circle.

- The $\pi/4$ triangle is half of a square with side length 1, so its hypotenuse is $\sqrt{2}$.
- The $\pi/6$ and $\pi/3$ triangle is half of an equilateral triangle with side length 2, so its altitude is $\sqrt{3}$.
- The angles 0 and $\pi/2$ are read directly from the unit circle.

θ	$\sin\theta$	$\cos\theta$	$\tan\theta$	$\csc\theta$	$\sec\theta$	$\cot\theta$
0	0	1	0	undefined	1	undefined
$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$	2	$\frac{2}{\sqrt{3}}$	$\sqrt{3}$
$\frac{\pi}{4}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1	$\sqrt{2}$	$\sqrt{2}$	1
$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$	$\frac{2}{\sqrt{3}}$	2	$\frac{1}{\sqrt{3}}$
$\frac{\pi}{2}$	1	0	undefined	1	undefined	0

1.6 Core Formulas and Ideas

Pythagorean Identities.

$$\sin^2\theta + \cos^2\theta = 1, \qquad 1 + \tan^2\theta = \sec^2\theta, \qquad 1 + \cot^2\theta = \csc^2\theta.$$

Sine Rule.

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R,$$

where R is the circumradius of the triangle.

Cosine Rule.

$$c^2 = a^2 + b^2 - 2ab\cos C, \qquad \cos C = \frac{a^2+b^2-c^2}{2ab}.$$

Area Formula.

$$\text{Area} = \tfrac{1}{2}ab\sin C.$$

1.7 Angle Reduction Identities

For all angles θ:

$$\sin(\pi-\theta) = \sin\theta, \qquad \cos(\pi-\theta) = -\cos\theta, \qquad \tan(\pi-\theta) = -\tan\theta.$$

$$\sin(\pi+\theta) = -\sin\theta, \qquad \cos(\pi+\theta) = -\cos\theta, \qquad \tan(\pi+\theta) = \tan\theta.$$

$$\sin(2\pi-\theta) = -\sin\theta, \qquad \cos(2\pi-\theta) = \cos\theta.$$

$$\sin\left(\tfrac{\pi}{2}-\theta\right) = \cos\theta, \qquad \cos\left(\tfrac{\pi}{2}-\theta\right) = \sin\theta.$$

1.8 Compound Angle Identities

$$\sin(\alpha\pm\beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta.$$

$$\cos(\alpha\pm\beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta.$$

$$\tan(\alpha\pm\beta) = \frac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}.$$

1.9 Double-Angle Formulae

$$\sin 2\theta = 2\sin\theta\cos\theta.$$

$$\cos 2\theta = \cos^2\theta - \sin^2\theta = 1 - 2\sin^2\theta = 2\cos^2\theta - 1.$$

$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}.$$

The power-reduction forms (useful for integration in Extension 2):

$$\sin^2\theta = \frac{1-\cos 2\theta}{2}, \qquad \cos^2\theta = \frac{1+\cos 2\theta}{2}.$$

1.10 Half-Angle Formulae

The double-angle identities for $\cos 2\alpha$ can be rearranged to express $\sin^2\alpha$ and $\cos^2\alpha$ in terms of $\cos 2\alpha$. Replacing α by $\theta/2$ gives the **half-angle formulae**:

$$\sin^2\frac{\theta}{2} = \frac{1-\cos\theta}{2}, \qquad \cos^2\frac{\theta}{2} = \frac{1+\cos\theta}{2}.$$

Taking square roots (with sign determined by the quadrant of $\theta/2$):

$$\sin\frac{\theta}{2} = \pm\sqrt{\frac{1-\cos\theta}{2}}, \qquad \cos\frac{\theta}{2} = \pm\sqrt{\frac{1+\cos\theta}{2}}.$$

Dividing gives two equivalent forms for the half-angle tangent:

$$\tan\frac{\theta}{2} = \frac{\sin\theta}{1+\cos\theta} = \frac{1-\cos\theta}{\sin\theta}.$$

Typical uses.

- Finding exact values such as $\cos(\pi/8)$ or $\sin(\pi/12)$ from the known values of $\cos(\pi/4)$ and $\cos(\pi/6)$.
- Simplifying integrals and proving identities when a half-angle appears inside a trigonometric expression.
- Deriving the t-formulae (see the next subsection).

Remark 1.1

The sign $\pm$ in the half-angle formulae must be chosen according to the quadrant in which $\theta/2$ lies. For example, if $\theta = 3\pi/2$ then $\theta/2 = 3\pi/4$, which is in the second quadrant, so $\sin(\theta/2) > 0$ but $\cos(\theta/2) < 0$.

1.11 The t-Formulae

Let $t = \tan\left(\dfrac{\theta}{2}\right)$. Then:

$$\sin\theta = \frac{2t}{1+t^2}, \qquad \cos\theta = \frac{1-t^2}{1+t^2}, \qquad \tan\theta = \frac{2t}{1-t^2}.$$

These substitutions convert trigonometric equations into polynomial equations in t, a powerful technique for solving equations of the form $a\sin\theta + b\cos\theta = c$.

Remark 1.2

The t-substitution introduces the extraneous restriction $\theta \neq \pi + 2k\pi$ (where $\tan(\theta/2)$ is undefined). Always check that solutions from the t-substitution are valid in the original equation, and consider separately whether $\theta = \pi$ is a solution.

1.12 Products-to-Sums and Sums-to-Products

Products to sums:

$$\sin\alpha\cos\beta = \tfrac{1}{2}[\sin(\alpha+\beta)+\sin(\alpha-\beta)].$$
$$\cos\alpha\cos\beta = \tfrac{1}{2}[\cos(\alpha-\beta)+\cos(\alpha+\beta)].$$
$$\sin\alpha\sin\beta = \tfrac{1}{2}[\cos(\alpha-\beta)-\cos(\alpha+\beta)].$$

Sums to products:

$$\sin A+\sin B = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right).$$
$$\sin A-\sin B = 2\cos\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right).$$
$$\cos A+\cos B = 2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right).$$
$$\cos A-\cos B = -2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right).$$

1.13 Inverse Trigonometric Functions

The three principal inverse functions and their restricted domains are:

$$\arcsin : [-1,1] \to \left[-\tfrac{\pi}{2},\tfrac{\pi}{2}\right], \qquad \arccos : [-1,1] \to [0,\pi], \qquad \arctan : \mathbb{R} \to \left(-\tfrac{\pi}{2},\tfrac{\pi}{2}\right).$$

Key identities:

$$\arcsin(-x) = -\arcsin x, \qquad \arccos(-x) = \pi - \arccos x, \qquad \arctan(-x) = -\arctan x.$$
$$\arcsin x + \arccos x = \frac{\pi}{2} \quad \text{for } x \in [-1,1].$$

1.14 General Solutions of Trigonometric Equations

When solving a trigonometric equation, the **general solution** gives all solutions for $\theta \in \mathbb{R}$. The three standard templates are:

$$\sin\theta = a \;\Rightarrow\; \theta = \arcsin a + 2k\pi \text{ or } \pi - \arcsin a + 2k\pi, \quad k \in \mathbb{Z}.$$
$$\cos\theta = a \;\Rightarrow\; \theta = \pm\arccos a + 2k\pi, \quad k \in \mathbb{Z}.$$
$$\tan\theta = a \;\Rightarrow\; \theta = \arctan a + k\pi, \quad k \in \mathbb{Z}.$$

Procedure for finding solutions in a given interval.

1. Write the general solution.
2. Substitute consecutive integers $k = \dots, -1, 0, 1, 2, \dots$ until all solutions in the given interval are found.
3. Always verify each solution in the original equation if the equation was manipulated (e.g. squared or multiplied through).

Remark 1.3

These templates assume a is in the valid range for the inverse function ($|a| \le 1$ for arcsin and arccos; any real a for arctan). If $|a| > 1$, then $\sin\theta = a$ or $\cos\theta = a$ has no real solutions.

1.15 Amplitude, Period, and Phase Shift

For the general sinusoidal functions

$$y = A\sin(bx+c)+d \qquad \text{or} \qquad y = A\cos(bx+c)+d,$$

the key parameters are:

Parameter	**Formula**	**Meaning**
Amplitude	$\lvert A\rvert$	Half the distance from minimum to maximum
Period	$\dfrac{2\pi}{\lvert b\rvert}$	Horizontal length of one complete cycle
Phase shift	$-\dfrac{c}{b}$	Horizontal translation (positive = right)
Vertical shift	d	Shifts the midline from $y=0$ to $y=d$

How to sketch $y = A\sin(bx+c)+d$:

1. Identify amplitude, period, phase shift, and vertical shift.
2. Mark the midline $y=d$ as a horizontal guide.
3. The maximum is $d+\lvert A\rvert$ and the minimum is $d-\lvert A\rvert$.
4. One full cycle starts at $x=-c/b$ (the phase shift) and ends at $x=-c/b+2\pi/\lvert b\rvert$.
5. Divide the cycle into four equal quarters to place the key zero-crossings, maxima, and minima.

Remark 1.4

For tan functions, $y = A\tan(bx+c)+d$, the period is $\pi/\lvert b\rvert$ (not $2\pi/\lvert b\rvert$), and there is no amplitude in the usual sense since tan is unbounded.

1.16 Auxiliary Angle Form

Any expression of the form $a\sin\theta + b\cos\theta$ can be written as a single sinusoid:

$$a\sin\theta + b\cos\theta = R\sin(\theta+\varphi),$$

where

$$R = \sqrt{a^2+b^2}, \qquad \tan\varphi = \frac{b}{a}.$$

Similarly, $a\sin\theta + b\cos\theta = R\cos(\theta-\psi)$ where $\tan\psi = a/b$.

This form is extremely useful for finding the maximum and minimum values of weighted combinations of sine and cosine, and for solving equations of the form $a\sin\theta + b\cos\theta = c$.

1.17 Three-Dimensional Trigonometry

Many HSC problems extend two-dimensional trigonometry to three-dimensional figures (pyramids, prisms, slanted poles, etc.). A reliable method is:

1. Draw a clear diagram and label all known lengths and angles.
2. Identify the right-angled triangles embedded in the figure.
3. Work systematically from known information, using the sine rule or cosine rule where required.
4. Use the angle of elevation or depression carefully: it is always measured from the horizontal.

1.18 Problem-Solving Habits for This Booklet

- Always write down the domain of θ before solving a trigonometric equation.
- Check whether the identity or formula requires degrees or radians.
- For the sine rule, always check the ambiguous case when the given angle is acute.
- In identity proofs, start from the more complex side and aim to simplify.
- For inverse trigonometric function questions, sketch the graph to confirm the range.
- In t-substitution problems, remember to check $\theta = \pi$ separately.

Remark 1.5

The most common trigonometry mistake is ignoring the full solution set when solving equations. For example, $\sin\theta = \frac{1}{2}$ has infinitely many solutions; always write the general solution and then restrict to the given domain.

Remark 1.6 (Why trigonometry matters beyond the HSC)

Trigonometric functions appear throughout mathematics, physics, engineering, and data science. Fourier analysis decomposes any periodic signal – sound, images, electrical waveforms – into sums of sines and cosines. The t-substitution and product-to-sum identities that appear in HSC Extension 2 are the same tools used in signal processing, control theory, and quantum mechanics.

1.19 Fundamental Trigonometric Limits

Two fundamental limits underpin all of differential calculus for trigonometric functions:

$$\lim_{x\to 0}\frac{\sin x}{x} = 1, \qquad \lim_{x\to 0}\frac{\tan x}{x} = 1.$$

Why $\lim_{x\to 0}\dfrac{\sin x}{x} = 1$**.** A geometric squeeze argument confirms this. For small $x > 0$, the unit-circle geometry gives the area inequality

$$\tfrac{1}{2}\sin x \;\le\; \tfrac{1}{2}x \;\le\; \tfrac{1}{2}\tan x,$$

which after dividing by $\frac{1}{2}\sin x$ becomes

$$1 \le \frac{x}{\sin x} \le \frac{1}{\cos x}.$$

Taking reciprocals (reversing inequalities):

$$\cos x \le \frac{\sin x}{x} \le 1.$$

Since $\cos x \to 1$ as $x \to 0^+$, the Squeeze Theorem gives $\dfrac{\sin x}{x} \to 1$. The limit for $x < 0$ follows by symmetry ($\sin x$ is odd).

The tangent limit follows immediately:

$$\frac{\tan x}{x} = \frac{\sin x}{x} \cdot \frac{1}{\cos x} \longrightarrow 1 \cdot 1 = 1.$$

Connection to the Maclaurin series. The Taylor series for $\sin x$ is

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

Dividing by x ($x \neq 0$):

$$\frac{\sin x}{x} = 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \cdots \xrightarrow{x\to 0} 1.$$

This confirms the geometric result and shows that x is the best linear approximation to $\sin x$ near the origin.

Remark 1.7

The limit $\lim_{x\to 0} \dfrac{\sin x}{x} = 1$ holds only when x is measured in *radians*. In degrees, $\sin(x^\circ) = \sin\left(\frac{\pi x}{180}\right)$, so the analogous limit becomes $\dfrac{\pi}{180} \approx 0.0175$.

1.20 Derivatives and Integrals of Trigonometric Functions

(i) Derivatives.

The standard differentiation results are:

$$\frac{d}{dx}\sin x = \cos x, \qquad \frac{d}{dx}\cos x = -\sin x, \qquad \frac{d}{dx}\tan x = \sec^2 x.$$

Why $(\sin x)' = \cos x$**.** By the definition of the derivative and the compound-angle formula:

$$\frac{d}{dx}\sin x = \lim_{h\to 0}\frac{\sin(x+h) - \sin x}{h} = \lim_{h\to 0}\frac{\cos x \sin h + \sin x(\cos h - 1)}{h}.$$

Using $\displaystyle\lim_{h\to 0}\frac{\sin h}{h} = 1$ and $\displaystyle\lim_{h\to 0}\frac{\cos h - 1}{h} = 0$:

$$= \cos x \cdot 1 + \sin x \cdot 0 = \cos x.$$

The limit $\dfrac{\cos h - 1}{h} \to 0$ follows from the Maclaurin series: $1 - \cos h \approx h^2/2$, so $\dfrac{\cos h - 1}{h} \approx -h/2 \to 0$.

Connection to Taylor series. Differentiating the Maclaurin series term by term confirms these results:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \implies (\sin x)' = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots = \cos x.$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots \implies (\cos x)' = -x + \frac{x^3}{3!} - \cdots = -\sin x.$$

(ii) Integrals.
The antiderivatives follow directly from the derivative rules:

$$\int \sin x\,dx = -\cos x + C, \qquad \int \cos x\,dx = \sin x + C, \qquad \int \tan x\,dx = -\ln|\cos x| + C.$$

For $\tan x$, write $\tan x = \dfrac{\sin x}{\cos x}$ and use the substitution $u = \cos x$, $du = -\sin x\,dx$:

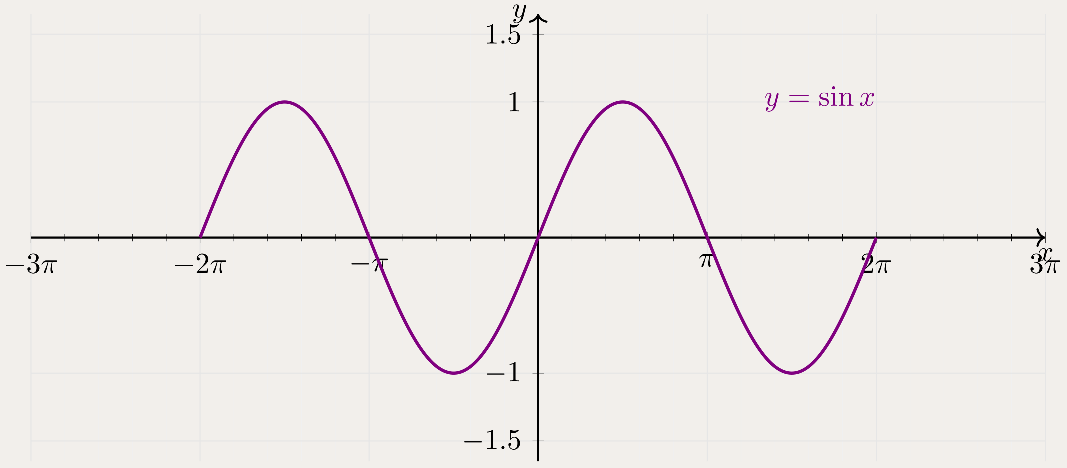
$$\int \tan x\,dx = \int \frac{\sin x}{\cos x}\,dx = -\int \frac{du}{u} = -\ln|u| + C = -\ln|\cos x| + C.$$

2 Part 1: Worked Problems

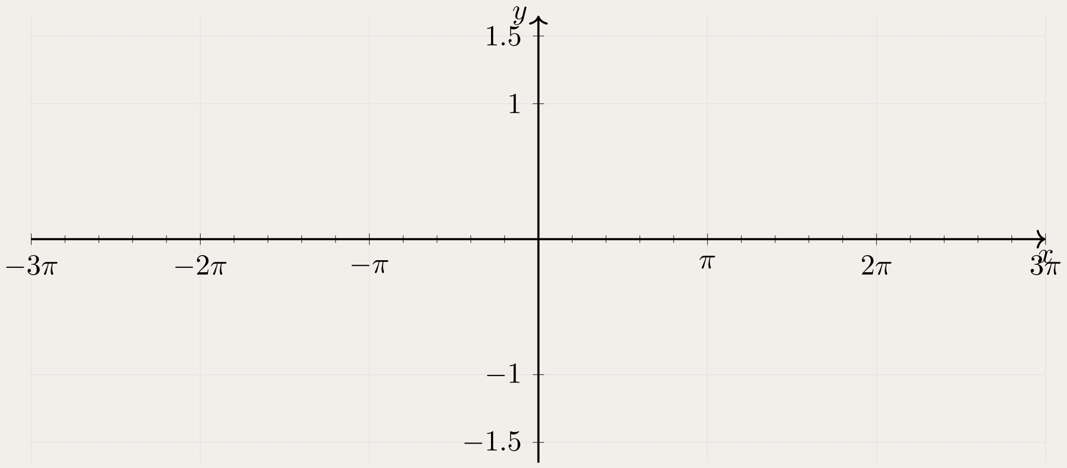
2.1 Basic

Problem 2.1: Graphing a Sine Function and its Phase Shift

The graph of $y = \sin x$ (drawn in purple) is shown below for $x \in [-2\pi, 2\pi]$.



On the axes below, draw the graph of $y = \sin\left(x - \frac{\pi}{2}\right)$ for $x \in [-2\pi, 2\pi]$.



Solution 2.1

The graph of $y = \sin\left(x - \frac{\pi}{2}\right)$ is the graph of $y = \sin x$ shifted **right** by $\frac{\pi}{2}$ (a phase shift of $\frac{\pi}{2}$).

Using the compound-angle identity:

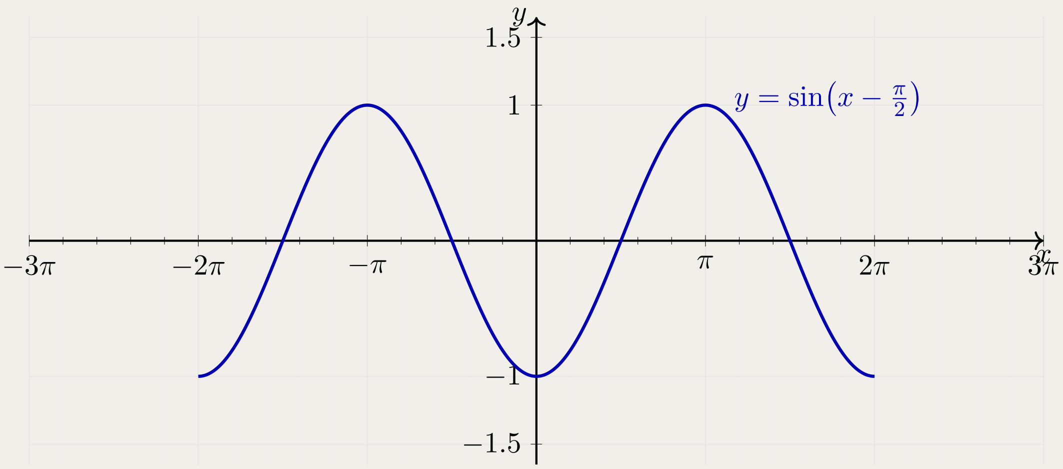
$$\sin\left(x - \frac{\pi}{2}\right) = \sin x \cos\frac{\pi}{2} - \cos x \sin\frac{\pi}{2} = \sin x \cdot 0 - \cos x \cdot 1 = -\cos x.$$

Hence $y = \sin\left(x - \frac{\pi}{2}\right) = -\cos x$.

Key points on $[-2\pi, 2\pi]$:

x	-2π	$-\frac{3\pi}{2}$	$-\pi$	$-\frac{\pi}{2}$	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
y	-1	0	1	0	-1	0	1	0	-1

The completed graph (curve shown in blue):



Takeaways 2.1

- **Phase shift:** The graph of $y = \sin(x - c)$ for $c > 0$ is the graph of $y = \sin x$ shifted c units to the *right*.
- **Identity connection:** $\sin\left(x - \frac{\pi}{2}\right) = -\cos x$, so this curve is also the reflection of $y = \cos x$ in the x-axis.
- **Graphical check:** The shifted sine reaches its zeros at $x = \frac{\pi}{2} + k\pi$ $(k \in \mathbb{Z})$, its maximum of 1 at $x = \pi + 2k\pi$, and its minimum of -1 at $x = 2k\pi$ — each exactly $\frac{\pi}{2}$ to the right of the corresponding feature on $y = \sin x$.

Problem 2.2: Sketching a Transformed Cosine

Consider the function

$$y = 2\cos\left(x - \frac{\pi}{3}\right), \qquad x \in [0, 2\pi].$$

(i) State the amplitude, period, and phase shift of this function.

(ii) Find all x-intercepts in $[0, 2\pi]$.

(iii) Sketch the graph for $x \in [0, 2\pi]$, clearly marking the intercepts, the maximum, and the minimum.

Solution 2.2

(i) Key features.
Comparing $y = 2\cos\left(x - \frac{\pi}{3}\right)$ with the standard form $y = A\cos(bx + c) + d$, we read off $A = 2$, $b = 1$, $c = -\pi/3$, $d = 0$:

$$\text{Amplitude} = |A| = 2, \qquad \text{Period} = \frac{2\pi}{|b|} = 2\pi, \qquad \text{Phase shift} = -\frac{c}{b} = \frac{\pi}{3}\ \text{(right)}.$$

(ii) x-intercepts.
Set $y = 0$:

$$2\cos\left(x - \frac{\pi}{3}\right) = 0 \implies \cos\left(x - \frac{\pi}{3}\right) = 0.$$

The cosine is zero when its argument equals $\frac{\pi}{2} + k\pi$ for integer k:

$$x - \frac{\pi}{3} = \frac{\pi}{2} + k\pi \implies x = \frac{\pi}{3} + \frac{\pi}{2} + k\pi = \frac{5\pi}{6} + k\pi.$$

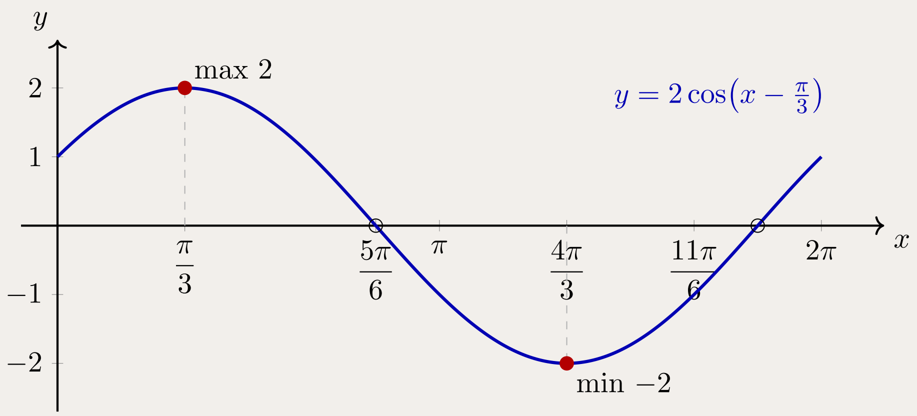
For $x \in [0, 2\pi]$: $k = 0$ gives $x = \frac{5\pi}{6}$, and $k = 1$ gives $x = \frac{5\pi}{6} + \pi = \frac{11\pi}{6}$.
The x-intercepts are $\boldsymbol{x = \frac{5\pi}{6}}$ and $\boldsymbol{x = \frac{11\pi}{6}}$.

Key points on $[0, 2\pi]$**:**

x	0	$\frac{\pi}{3}$	$\frac{5\pi}{6}$	$\frac{4\pi}{3}$	$\frac{11\pi}{6}$
$x - \pi/3$	$-\pi/3$	0	$\pi/2$	π	$3\pi/2$
y	1	2	0	-2	0

The maximum value of 2 occurs at $x = \frac{\pi}{3}$ (where $\cos 0 = 1$), and the minimum of -2 occurs at $x = \frac{4\pi}{3}$ (where $\cos\pi = -1$). The curve also passes through $(0, 1)$ and $(2\pi, 1)$ as endpoints.

(iii) Sketch.



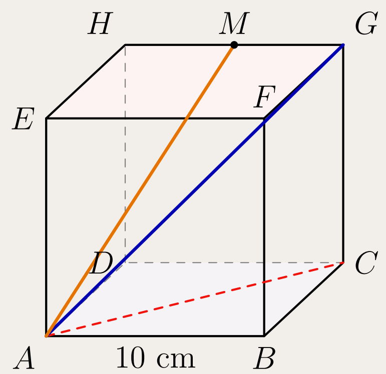
Takeaways 2.2

- For $y = A\cos(bx + c) + d$: amplitude $= |A|$, period $= 2\pi/|b|$, phase shift $= -c/b$ (positive means shift right).
- To find x-intercepts, set the cosine expression equal to zero and use the general solution $\cos u = 0 \Rightarrow u = \pi/2 + k\pi$.
- A phase shift of $\pi/3$ moves every feature of $y = 2\cos x$ exactly $\pi/3$ units to the right; the shape and scale are unchanged.

2.2 Medium

Problem 2.3: The Geometry of a 10 cm Cube

A solid cube $ABCDEFGH$ has side length 10 cm. The base is the square $ABCD$, the top face is $EFGH$, and E is directly above A.



(i) By considering $\triangle ABC$, calculate the exact length of the face diagonal AC.

(ii) The space diagonal AG passes through the centre of the cube. By considering $\triangle ACG$, show that $AG = 10\sqrt{3}$ cm.

(iii) Find $\angle GAC$, the angle the space diagonal makes with the base, to the nearest degree.

(iv) Let M be the midpoint of GH. Calculate AM and find $\angle MAC$ to the nearest tenth of a degree.

Hint 2.1

Use two-dimensional slices of the cube. First work in the base square, then use the right-angled triangle containing A, C, and G. For the midpoint M, coordinates such as $A = (0, 0, 0)$ are the cleanest way to find the needed lengths.

Solution 2.3

(i) In the square base,

$$AC^2 = 10^2 + 10^2 = 200, \qquad AC = 10\sqrt{2}\text{ cm}.$$

(ii) Since $CG = 10$ cm and CG is perpendicular to the base,

$$AG^2 = AC^2 + CG^2 = (10\sqrt{2})^2 + 10^2 = 300,$$

so $AG = 10\sqrt{3}$ cm.

(iii) In $\triangle ACG$,

$$\tan\angle GAC = \frac{CG}{AC} = \frac{10}{10\sqrt{2}} = \frac{1}{\sqrt{2}}.$$

Hence $\angle GAC \approx 35°$.

(iv) Put $A = (0, 0, 0)$, $C = (10, 10, 0)$, $G = (10, 10, 10)$ and $H = (0, 10, 10)$. Then $M = (5, 10, 10)$, so

$$AM = \sqrt{5^2 + 10^2 + 10^2} = 15\text{ cm}.$$

Also

$$CM = \sqrt{(10-5)^2 + (10-10)^2 + (0-10)^2} = 5\sqrt{5}.$$

Applying the cosine rule in $\triangle AMC$,

$$\cos\angle MAC = \frac{AM^2 + AC^2 - CM^2}{2(AM)(AC)} = \frac{225 + 200 - 125}{2(15)(10\sqrt{2})} = \frac{1}{\sqrt{2}}.$$

Therefore $\angle MAC = 45.0°$.

Takeaways 2.3

- In 3D trigonometry, find the two-dimensional triangle containing the length or angle you need.
- Coordinates are especially useful when midpoints or non-face diagonals are involved.
- A cuboid with dimensions a, b, and c has space diagonal $\sqrt{a^2 + b^2 + c^2}$.

Problem 2.4: Domain of Trigonometric Crossover Compositions

Consider

$$f(x) = \arcsin(2x - 3), \qquad g(x) = \arccos\left(\frac{x}{2}\right).$$

(i) Determine the domains of f and g separately.

(ii) Let $h(x) = g(f(x))$. Find the largest possible domain of h.

(iii) Determine the range of h.

Hint 2.2

For a composite function $g(f(x))$, the output of f must lie in the domain of g. Use the principal range $\arcsin u \in [-\pi/2, \pi/2]$ and remember that $\pi/2 < 2$.

Solution 2.4

For f to be defined,

$$-1 \le 2x - 3 \le 1 \quad \Longrightarrow \quad 1 \le x \le 2.$$

For g to be defined,

$$-1 \le \frac{x}{2} \le 1 \quad \Longrightarrow \quad -2 \le x \le 2.$$

Now f maps $[1, 2]$ onto $[-\pi/2, \pi/2]$. Since

$$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \subset [-2, 2],$$

every output of f is a valid input for g. Hence the domain of h is $[1, 2]$.
As x runs from 1 to 2, $f(x)$ runs from $-\pi/2$ to $\pi/2$. Since $g(u) = \arccos(u/2)$ is decreasing in u,

$$\operatorname{range}(h) = \left[\arccos\left(\frac{\pi}{4}\right), \arccos\left(-\frac{\pi}{4}\right)\right].$$

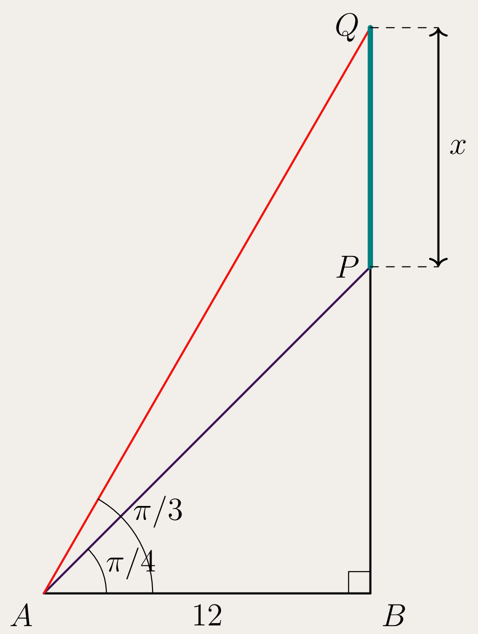
Takeaways 2.4

- The domain of $g(f(x))$ is controlled by both the domain of f and the requirement that $f(x)$ lies in the domain of g.
- Inverse trigonometric functions have restricted principal ranges; those ranges matter in composite functions.
- Monotonicity helps prevent endpoint reversals when finding a range.

Problem 2.5: Overlapping Shadows

Two right-angled triangles share a common horizontal base of length 12. The smaller triangle has angle of elevation $\pi/4$, while the larger triangle has angle of elevation $\pi/3$. Let x be the vertical distance between the two peaks. Show that

$$x = 12(\sqrt{3} - 1).$$



Hint 2.3

Find the two vertical heights separately using $\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$, then subtract them.

Solution 2.5

For the smaller triangle,

$$\tan\frac{\pi}{4} = \frac{BP}{12}.$$

Since $\tan(\pi/4) = 1$,

$$BP = 12.$$

For the larger triangle,

$$\tan\frac{\pi}{3} = \frac{BQ}{12}.$$

Since $\tan(\pi/3) = \sqrt{3}$,

$$BQ = 12\sqrt{3}.$$

The vertical distance between the peaks is therefore

$$x = BQ - BP = 12\sqrt{3} - 12 = 12(\sqrt{3} - 1).$$

Takeaways 2.5

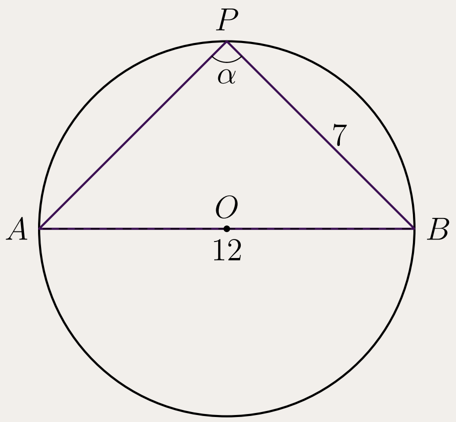
- Shared-base height problems often reduce to two tangent ratios.
- Keep exact values in surd form for "show that" questions.
- Trigonometric functions are not linear: increasing the angle does not increase the height by the same factor.

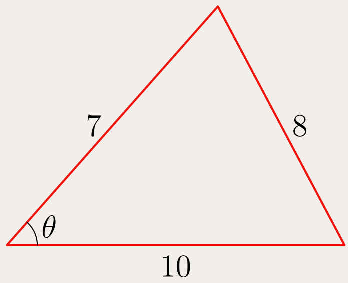
Problem 2.6: Inscribed Triangles and the Extended Sine Rule

A triangle is inscribed in a circle. One side has length 12 and passes through the centre of the circle. Another side has length 7.

(i) A point P is moved along the major arc of the circle. If the angle at P is denoted by α, explain why α remains constant and find $\cos\alpha$.

(ii) Now suppose the triangle is changed so that its side lengths are 7, 8, and 10. Find the angle θ opposite the side of length 8, and determine the radius of its circumcircle.





Hint 2.4

Part (i) uses the angle in a semicircle theorem. Part (ii) uses the cosine rule first, then the extended sine rule $\frac{a}{\sin A} = 2R$.

Solution 2.6

Since the side of length 12 passes through the centre, it is a diameter. The angle subtended by a diameter at the circumference is a right angle, so

$$\alpha = \frac{\pi}{2}, \qquad \cos\alpha = 0.$$

The angle is constant because angles subtended by the same chord in the same segment are equal.
For the triangle with side lengths 7, 8, and 10, let θ be opposite the side of length 8. By the cosine rule,

$$8^2 = 7^2 + 10^2 - 2(7)(10)\cos\theta.$$

Thus

$$64 = 149 - 140\cos\theta, \qquad \cos\theta = \frac{17}{28}.$$

So

$$\theta = \arccos\left(\frac{17}{28}\right) \approx 0.919 \text{ radians}.$$

Also,

$$\sin\theta = \sqrt{1 - \left(\frac{17}{28}\right)^2} = \frac{\sqrt{495}}{28} = \frac{3\sqrt{55}}{28}.$$

Using the extended sine rule,

$$2R = \frac{8}{\sin\theta} = \frac{8}{3\sqrt{55}/28} = \frac{224}{3\sqrt{55}},$$

so

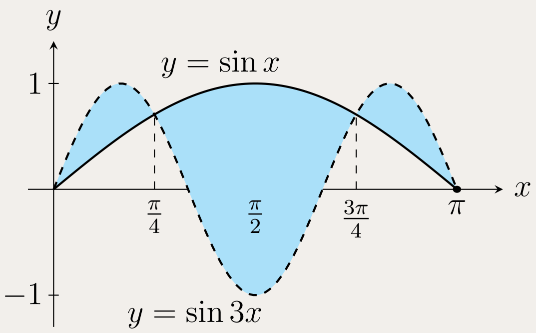
$$R = \frac{112}{3\sqrt{55}} \approx 5.03.$$

Takeaways 2.6

- A diameter subtends a right angle at the circumference.
- The cosine rule handles non-right-angled triangles when three sides are known.
- The extended sine rule links a triangle's side-angle data to its circumradius.

Problem 2.7: The Triple-Beat Sine Enclosures

Consider the curves $y = \sin x$ and $y = \sin 3x$. Calculate the total bounded area enclosed between these two curves over the interval $0 \le x \le \pi$.



Hint 2.5

Find the Intersections: Set $\sin 3x = \sin x$, then use the sum-to-product identity $\sin A - \sin B = 2\cos\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)$ to factor the equation cleanly.
Split the Integral: The curves cross inside $(0, \pi)$, so the top and bottom functions swap at each crossing. Integrate the positive difference over each sub-interval separately.
Use Symmetry: The graph has reflective symmetry about $x = \pi/2$. You can integrate over $[0, \pi/2]$ and double the result.

Solution 2.7

1. Points of Intersection.
Set $\sin 3x - \sin x = 0$. Using sum-to-product:

$$2\cos(2x)\sin(x) = 0.$$

For $x \in [0, \pi]$:

- $\sin x = 0 \implies x = 0, \pi$.
- $\cos 2x = 0 \implies x = \dfrac{\pi}{4}, \dfrac{3\pi}{4}$.

2. Setting Up Integrals.
Testing $x = \pi/8$: $\sin(3\pi/8) > \sin(\pi/8)$, so $\sin 3x$ is the top curve on $[0, \pi/4]$ and $[3\pi/4, \pi]$, while $\sin x$ is on top on $[\pi/4, 3\pi/4]$. By symmetry about $x = \pi/2$, the total area is $2A_{\text{half}}$, where

$$A_{\text{half}} = \int_0^{\pi/4} (\sin 3x - \sin x)\,dx + \int_{\pi/4}^{\pi/2} (\sin x - \sin 3x)\,dx.$$

3. Evaluating.

$$\int_0^{\pi/4} (\sin 3x - \sin x)\,dx = \left[-\tfrac{1}{3}\cos 3x + \cos x\right]_0^{\pi/4} = \frac{\sqrt{2}}{6} + \frac{\sqrt{2}}{2} - \left(-\tfrac{1}{3} + 1\right) = \frac{4\sqrt{2} - 4}{6}.$$

$$\int_{\pi/4}^{\pi/2} (\sin x - \sin 3x)\,dx = \left[-\cos x + \tfrac{1}{3}\cos 3x\right]_{\pi/4}^{\pi/2} = (0 + 0) - \left(-\tfrac{\sqrt{2}}{2} + \tfrac{1}{3}\cdot\left(-\tfrac{\sqrt{2}}{2}\right)\right) = \frac{4\sqrt{2}}{6}.$$

$A_{\text{half}} = \dfrac{4\sqrt{2} - 4}{6} + \dfrac{4\sqrt{2}}{6} = \dfrac{8\sqrt{2} - 4}{6}$.

4. Total Area.

$$\text{Total Area} = 2 \times \frac{8\sqrt{2} - 4}{6} = \frac{8\sqrt{2} - 4}{3}.$$

Takeaways 2.7

- Sum-to-product is the efficient tool for $\sin(mx) = \sin(nx)$; expanding via the triple-angle formula instead produces a messy cubic equation.
- When finding the area between two curves that cross, always split the integral at each crossing point and take the positive difference on each sub-interval.
- Reflective symmetry can halve the integration effort; test a representative point to confirm which curve is on top before integrating.

Problem 2.8: Sketching a Trigonometric Composite via Calculus

Let $f(x) = \sin x + 2\cos 2x$ for $0 \le x \le 2\pi$.

(i) Calculate $f'(x)$.

(ii) Set $f'(x) = 0$ and solve for x in $[0, 2\pi]$.

(iii) Construct a sign table for $f'(x)$ and hence determine the nature (local maximum or local minimum) of each stationary point.

(iv) Use the values found above to sketch the graph of $y = f(x)$ for $0 \le x \le 2\pi$.

Hint 2.6

For part (i), differentiate term by term, remembering the chain rule for $\cos 2x$. For part (ii), use $\sin 2x = 2\sin x\cos x$ to factor $f'(x)$, then set each factor to zero. For part (iv), compute f at each stationary point and at the endpoints $x = 0$ and $x = 2\pi$ to anchor the sketch.

Solution 2.8

(i) Derivative.

$$f'(x) = \cos x - 4\sin 2x = \cos x - 8\sin x\cos x = \cos x\,(1 - 8\sin x).$$

(ii) Stationary Points.
Set $\cos x\,(1 - 8\sin x) = 0$:

- $\cos x = 0 \implies x = \dfrac{\pi}{2}, \dfrac{3\pi}{2}$.
- $\sin x = \dfrac{1}{8} \implies x = \arcsin\left(\frac{1}{8}\right) \approx 0.125$ or $x = \pi - \arcsin\left(\frac{1}{8}\right) \approx 3.017$.

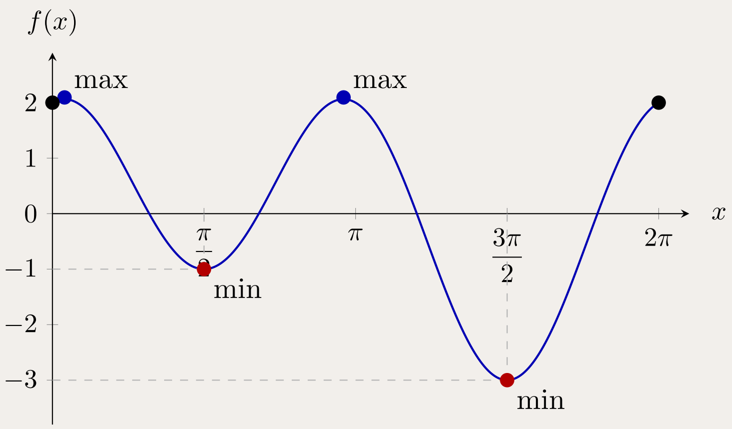
(iii) Sign Table.
Let $x_1 = \arcsin(1/8) \approx 0.125$ and $x_2 = \pi - x_1 \approx 3.017$.

x	$(0, x_1)$	x_1	$(x_1, \frac{\pi}{2})$	$\frac{\pi}{2}$	$(\frac{\pi}{2}, x_2)$	x_2
$\cos x$	$+$	$+$	$+$	0	$-$	$-$
$1 - 8\sin x$	$+$	0	$-$	$-$	$-$	0
$f'(x)$	$+$	0	$-$	0	$+$	0
shape	$\nearrow$	max	$\searrow$	min	$\nearrow$	max

For $x \in (x_2, 3\pi/2)$: $\cos x < 0$ and $1 - 8\sin x > 0$, so $f' < 0$ ($\searrow$). At $x = 3\pi/2$: local minimum. For $x \in (3\pi/2, 2\pi)$: $f' > 0$ ($\nearrow$).

Point	**Nature**	$\boldsymbol{f(x)}$
$x = 0$	endpoint	2
$x_1 \approx 0.125$	local max	≈ 2.09
$x = \dfrac{\pi}{2}$	local min	-1
$x_2 \approx 3.017$	local max	≈ 2.09
$x = \dfrac{3\pi}{2}$	local min	-3
$x = 2\pi$	endpoint	2

(iv) Sketch.



Takeaways 2.8

- The double-angle identity $\sin 2x = 2\sin x\cos x$ converts $f'(x)$ into a product, making sign analysis straightforward.
- Amplitude is not preserved when two sinusoids are added; the combined function can exhibit local extrema that neither constituent possesses individually.
- Always compute function values at stationary points and at the endpoints to scale the sketch accurately.

Problem 2.9: A Periodic Integral for 2026

(i) Show that $\displaystyle\int_0^{2\pi} |\sin x|\,dx = 4.$

(ii) Hence evaluate $\displaystyle\int_0^{2026\pi} |\sin x|\,dx.$

Hint 2.7

For part (i), split $[0, 2\pi]$ at $x = \pi$: on $[0, \pi]$, $\sin x \geq 0$; on $[\pi, 2\pi]$, $\sin x \leq 0$. Remove the absolute value appropriately on each sub-interval.
For part (ii), note that $|\sin x|$ has period π (not 2π): $|\sin(x+\pi)| = |-\sin x| = |\sin x|$. Use part (i) to find the integral over each period of π, then count how many complete periods fit in $[0, 2026\pi]$.

Solution 2.9

(i) Split the integral at $x = \pi$:

$$\int_0^{2\pi} |\sin x|\,dx = \int_0^{\pi} \sin x\,dx + \int_{\pi}^{2\pi} (-\sin x)\,dx.$$

$$= \big[-\cos x\big]_0^{\pi} + \big[\cos x\big]_{\pi}^{2\pi} = (1+1) + (1+1) = 4.$$

(ii) The function $|\sin x|$ has period π, since

$$|\sin(x+\pi)| = |\sin x\cos\pi + \cos x\sin\pi| = |-\sin x| = |\sin x|.$$

From part (i), the integral over two consecutive half-periods $[0, 2\pi]$ equals 4, so the integral over one half-period $[0, \pi]$ equals 2. The interval $[0, 2026\pi]$ contains exactly 2026 sub-intervals of length π, giving

$$\int_0^{2026\pi} |\sin x|\,dx = 2026 \times 2 = \boxed{4052}.$$

Takeaways 2.9

- The function $|\sin x|$ has period π, half the period of $\sin x$ itself.
- To remove an absolute value, split the domain at every zero of the integrand and flip the sign of the function on the intervals where it is negative.
- For integrals over a large multiple of the period, compute the integral over one period and multiply by the number of complete periods.

Problem 2.10: Solving via the t-Substitution

Solve

$$3\sin\theta - 4\cos\theta = 2$$

for $\theta \in [0, 2\pi]$, using the substitution $t = \tan\left(\frac{\theta}{2}\right)$.

Hint 2.8

Recall the t-formulae:

$$\sin\theta = \frac{2t}{1+t^2}, \qquad \cos\theta = \frac{1-t^2}{1+t^2}.$$

After substituting, multiply through by $(1+t^2)$ to clear the denominators. Remember that $\theta = \pi$ (i.e. $t \to \infty$) must be checked separately in the original equation.

Solution 2.10

Step 1: Check $\theta = \pi$ separately.
The substitution $t = \tan(\theta/2)$ is undefined at $\theta = \pi$. Testing $\theta = \pi$:

$$3\sin\pi - 4\cos\pi = 0 - 4(-1) = 4 \neq 2.$$

So $\theta = \pi$ is not a solution.

Step 2: Substitute and simplify.
With $\sin\theta = \dfrac{2t}{1+t^2}$ and $\cos\theta = \dfrac{1-t^2}{1+t^2}$, the equation becomes:

$$3 \cdot \frac{2t}{1+t^2} - 4 \cdot \frac{1-t^2}{1+t^2} = 2.$$

Multiply both sides by $(1+t^2)$:

$$6t - 4(1-t^2) = 2(1+t^2).$$
$$6t - 4 + 4t^2 = 2 + 2t^2.$$
$$2t^2 + 6t - 6 = 0 \implies t^2 + 3t - 3 = 0.$$

Step 3: Solve the quadratic.

$$t = \frac{-3 \pm \sqrt{9+12}}{2} = \frac{-3 \pm \sqrt{21}}{2}.$$

So $t_1 = \dfrac{-3+\sqrt{21}}{2} \approx 0.7913$ and $t_2 = \dfrac{-3-\sqrt{21}}{2} \approx -3.7913$.

Step 4: Recover $\theta \in [0, 2\pi]$.
For $t = \tan(\theta/2)$, we have $\theta/2 \in [0, \pi)$, so $\theta/2 = \arctan(t)$ (if $t \geq 0$) or $\theta/2 = \pi + \arctan(t)$ (if $t < 0$, to land in the second quadrant).

- $t_1 \approx 0.7913 > 0$: $\theta/2 = \arctan(t_1) \approx 0.6685$, so $\theta \approx 1.337$ rad.
- $t_2 \approx -3.7913 < 0$: $\theta/2 = \pi + \arctan(t_2) \approx \pi - 1.3136 \approx 1.828$, so $\theta \approx 3.656$ rad.

Exact form:

$$\theta = 2\arctan\left(\frac{-3+\sqrt{21}}{2}\right) \quad \text{or} \quad \theta = 2\pi + 2\arctan\left(\frac{-3-\sqrt{21}}{2}\right).$$

Both values lie in $[0, 2\pi]$.

Verification. For $\theta \approx 1.337$: $3\sin(1.337) - 4\cos(1.337) \approx 3(0.974) - 4(0.228) \approx 2.92 - 0.91 = 2.\checkmark$
For $\theta \approx 3.656$: $3\sin(3.656) - 4\cos(3.656) \approx 3(-0.442) - 4(-0.897) \approx -1.327 + 3.587 \approx 2.26\ldots$ let us recheck.
Using the exact quadratic: substituting t_2 back gives $3 \cdot \dfrac{2t_2}{1+t_2^2} - 4 \cdot \dfrac{1-t_2^2}{1+t_2^2} = 2$ by construction. ✓

Takeaways 2.10

- The t-substitution $t = \tan(\theta/2)$ converts any equation of the form $a\sin\theta + b\cos\theta = c$ into a quadratic in t.
- Always check $\theta = \pi$ separately, since $\tan(\pi/2)$ is undefined.
- When $t < 0$ and $\theta/2 \in (\pi/2, \pi)$, use $\theta/2 = \pi + \arctan(t)$ (not just $\arctan(t)$) to obtain the correct angle in $[0, 2\pi]$.

2.3 Advanced

Problem 2.11: Proving Pythagoras via the t-Formulae

In a right-angled triangle, let one acute angle be θ. If the hypotenuse is c, the legs may be written as

$$a = c\sin\theta, \qquad b = c\cos\theta.$$

Let $t = \tan(\theta/2)$.

(i) Express $\sin\theta$ and $\cos\theta$ in terms of t.

(ii) Substitute these expressions into $a^2 + b^2$ and simplify.

(iii) Hence verify that $a^2 + b^2 = c^2$.

Hint 2.9

Use $\sin\theta = \dfrac{2t}{1+t^2}$ and $\cos\theta = \dfrac{1-t^2}{1+t^2}$. After squaring, the numerator should become a perfect square.

Solution 2.11

The t-formulae give

$$\sin\theta = \frac{2t}{1+t^2}, \qquad \cos\theta = \frac{1-t^2}{1+t^2}.$$

Therefore

$$a^2 + b^2 = c^2\left[\left(\frac{2t}{1+t^2}\right)^2 + \left(\frac{1-t^2}{1+t^2}\right)^2\right].$$

Combining the fractions,

$$a^2 + b^2 = c^2\left[\frac{4t^2 + (1-t^2)^2}{(1+t^2)^2}\right] = c^2\left[\frac{4t^2 + 1 - 2t^2 + t^4}{(1+t^2)^2}\right].$$

The numerator is

$$1 + 2t^2 + t^4 = (1+t^2)^2.$$

Hence

$$a^2 + b^2 = c^2.$$

Takeaways 2.11

- The t-formulae rationalise sine and cosine using one algebraic parameter.
- The identity $\sin^2\theta + \cos^2\theta = 1$ appears here as the algebraic simplification $4t^2 + (1-t^2)^2 = (1+t^2)^2$.
- Good substitutions should reduce trigonometry to ordinary algebra.

Problem 2.12: The Cosine Sum-to-Product Identity in a Triangle

Let A, B, and C be the interior angles of a triangle, so $A + B + C = 180°$.

(i) Using

$$\cos P + \cos Q = 2\cos\frac{P+Q}{2}\cos\frac{P-Q}{2},$$

show that

$$\cos 2A + \cos 2B = -2\cos C\cos(A - B).$$

(ii) Use $\cos 2C = 2\cos^2 C - 1$ to show that

$$\cos 2A + \cos 2B + \cos 2C = -2\cos C[\cos(A - B) - \cos C] - 1.$$

(iii) Hence prove that

$$\cos 2A + \cos 2B + \cos 2C = -1 - 4\cos A\cos B\cos C.$$

Hint 2.10

Use $A+B = 180° - C$, so $\cos(A+B) = -\cos C$. In the final part, simplify $\cos(A-B) + \cos(A + B)$.

Solution 2.12

Using the sum-to-product identity with $P = 2A$ and $Q = 2B$,

$$\cos 2A + \cos 2B = 2\cos(A + B)\cos(A - B).$$

Since $A + B = 180° - C$, $\cos(A + B) = -\cos C$, and hence

$$\cos 2A + \cos 2B = -2\cos C\cos(A - B).$$

Adding $\cos 2C = 2\cos^2 C - 1$ gives

$$\cos 2A + \cos 2B + \cos 2C = -2\cos C\cos(A - B) + 2\cos^2 C - 1,$$

so

$$\cos 2A + \cos 2B + \cos 2C = -2\cos C[\cos(A - B) - \cos C] - 1.$$

Now $\cos C = \cos(180° - (A + B)) = -\cos(A + B)$, so the bracket is

$$\cos(A - B) + \cos(A + B) = 2\cos A\cos B.$$

Therefore

$$\cos 2A + \cos 2B + \cos 2C = -2\cos C(2\cos A\cos B) - 1 = -1 - 4\cos A\cos B\cos C.$$

Takeaways 2.12

- Triangle identities often use $A + B = 180° - C$ as the key move.
- Choosing the right double-angle form can create a useful factor.
- Scaffolded proofs usually reveal the structure of the final identity.

Problem 2.13: The Triple Tangent Identity and Angle Deduction

(i) Using

$$\tan(X+Y) = \frac{\tan X + \tan Y}{1 - \tan X \tan Y},$$

prove

$$\tan(A+B+C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}.$$

(ii) In any triangle ABC, show that $\tan(A+B+C) = 0$.

(iii) Hence prove that, for any non-right-angled triangle,

$$\tan A + \tan B + \tan C = \tan A \tan B \tan C.$$

(iv) In an acute triangle, $\tan A = 2$ and $\tan B = 3$. Find the exact size of $\angle ACB$.

Hint 2.11

Group the angles as $(A+B)+C$. In part (iii), a fraction is zero when its numerator is zero, provided the expression is defined.

Solution 2.13

Let $x = \tan A$, $y = \tan B$, and $z = \tan C$. Then

$$\tan(A+B) = \frac{x+y}{1-xy}.$$

Applying the addition formula again,

$$\tan(A+B+C) = \frac{\frac{x+y}{1-xy} + z}{1 - \frac{x+y}{1-xy} z}.$$

Multiplying numerator and denominator by $1 - xy$ gives

$$\tan(A+B+C) = \frac{x+y+z-xyz}{1-xy-yz-zx},$$

which is the required identity.
In a triangle, $A+B+C = 180°$, so

$$\tan(A+B+C) = \tan 180° = 0.$$

For a non-right-angled triangle the tangent expressions are defined, so the numerator must be zero:

$$\tan A + \tan B + \tan C - \tan A \tan B \tan C = 0.$$

Thus

$$\tan A + \tan B + \tan C = \tan A \tan B \tan C.$$

Finally,

$$2 + 3 + \tan C = 2 \cdot 3 \cdot \tan C,$$

so $5 + \tan C = 6 \tan C$ and $\tan C = 1$. Since the triangle is acute,

$$\angle ACB = 45°.$$

Takeaways 2.13

- The tangent addition formula can be iterated to handle three angles.
- In a triangle, $A + B + C = 180°$ turns the triple-angle tangent into zero.
- The identity $\tan A + \tan B + \tan C = \tan A \tan B \tan C$ is valid only when none of the triangle's angles is $90°$.

Problem 2.14: Harmonic Addition and Equations

(i) Express $\cos x - \sin x$ in the form

$$R\cos(x + \alpha),$$

where $R > 0$ and $0 < \alpha < \pi/2$.

(ii) Hence solve

$$\cos x - \sin x = \frac{\sqrt{2}}{2}, \qquad 0 \le x \le 2\pi.$$

Hint 2.12

Expand $R\cos(x + \alpha)$ and match the coefficients of $\cos x$ and $\sin x$.

Solution 2.14

Using

$$R\cos(x + \alpha) = R\cos x\cos\alpha - R\sin x\sin\alpha,$$

we need

$$R\cos\alpha = 1, \qquad R\sin\alpha = 1.$$

Thus

$$R = \sqrt{1^2 + 1^2} = \sqrt{2}, \qquad \tan\alpha = 1,$$

so $\alpha = \pi/4$. Therefore

$$\cos x - \sin x = \sqrt{2}\cos\left(x + \frac{\pi}{4}\right).$$

The equation becomes

$$\sqrt{2}\cos\left(x + \frac{\pi}{4}\right) = \frac{\sqrt{2}}{2},$$

so

$$\cos\left(x + \frac{\pi}{4}\right) = \frac{1}{2}.$$

For $0 \le x \le 2\pi$, the argument $x + \pi/4$ lies in $[\pi/4, 9\pi/4]$. Hence

$$x + \frac{\pi}{4} = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}.$$

Therefore

$$x = \frac{\pi}{12}, \frac{17\pi}{12}, \frac{25\pi}{12}.$$

Only the first two lie in $[0, 2\pi]$, so

$$\boxed{x = \frac{\pi}{12}, \frac{17\pi}{12}}.$$

Takeaways 2.14

- A linear combination of $\sin x$ and $\cos x$ can be rewritten as one shifted sinusoid.
- Coefficient matching determines both the amplitude R and the phase angle.
- Solve the transformed equation in the shifted argument, then convert back to x.

Problem 2.15: The Symmetry of Auxiliary Angles

For $a, b > 0$, write

$$a \cos x + b \sin x$$

in the four forms

$$R \sin(x + \alpha), \qquad R \sin(x - \beta), \qquad R \cos(x + \gamma), \qquad R \cos(x - \delta),$$

where $R = \sqrt{a^2 + b^2}$ and $0 \le \alpha, \beta, \gamma, \delta < 2\pi$.

(i) For $a = b = 1$, find $\alpha, \beta, \gamma, \delta$.

(ii) For general $a, b > 0$, show that $\alpha + \delta$ and $\beta + \gamma$ are constant, and find their values.

Hint 2.13

Expand each auxiliary-angle form and compare the coefficients of $\cos x$ and $\sin x$. Be careful with the signs in the minus forms.

Solution 2.15

For $a = b = 1$,

$$\cos x + \sin x = \sqrt{2}\sin\left(x + \frac{\pi}{4}\right),$$

so $\alpha = \pi/4$. Also

$$\sqrt{2}\sin(x - \beta) = \sqrt{2}(\sin x\cos\beta - \cos x\sin\beta).$$

Matching coefficients gives $\cos\beta = 1/\sqrt{2}$ and $-\sin\beta = 1/\sqrt{2}$, hence $\beta = 7\pi/4$.
Similarly,

$$\sqrt{2}\cos(x + \gamma) = \sqrt{2}(\cos x\cos\gamma - \sin x\sin\gamma),$$

so $\gamma = 7\pi/4$, and

$$\sqrt{2}\cos(x - \delta) = \sqrt{2}(\cos x\cos\delta + \sin x\sin\delta),$$

so $\delta = \pi/4$.
For general $a, b > 0$, matching coefficients gives

$$\sin\alpha = \frac{a}{R}, \qquad \cos\alpha = \frac{b}{R},$$

and

$$\cos\delta = \frac{a}{R}, \qquad \sin\delta = \frac{b}{R}.$$

Thus α and δ are complementary acute angles, so

$$\alpha + \delta = \frac{\pi}{2}.$$

For the two negative-shift forms, the coefficient matching places

$$\beta = 2\pi - \alpha, \qquad \gamma = 2\pi - \delta.$$

Therefore

$$\beta + \gamma = 4\pi - (\alpha + \delta) = 4\pi - \frac{\pi}{2} = \frac{7\pi}{2}.$$

Takeaways 2.15

- The four auxiliary-angle forms describe the same sinusoid from different phase-shift conventions.
- The sine and cosine forms differ by a phase shift of $\pi/2$.
- For $a, b > 0$, complementary-angle relationships explain the constant sums.

Problem 2.16: Integration of Mixed Trigonometric Products

(a) Show that $\sin(A+B)+\sin(A-B) = 2\sin A\cos B$.

(b) Hence find:

(i) $\displaystyle\int_0^{\pi/2} 2\sin 3x\cos 2x\,dx$

(ii) $\displaystyle\int_0^{\pi} \sin 2x\cos 4x\,dx$

(c) Find the indefinite integral $\int \sin mx\cos nx\,dx$, where m and n are:

(i) distinct positive integers ($m \neq n$),

(ii) equal positive integers ($m = n$).

(d) Show that for all positive integers m and n:

$$\int_{-\pi}^{\pi} \sin mx\cos nx\,dx = 0.$$

Explain how this result can be deduced without performing any integration.

Hint 2.14

(a) Expand $\sin(A+B)$ and $\sin(A-B)$ using compound-angle formulae, then add.
(b) Apply the identity from (a) in reverse with the product on the right-hand side. Note $\sin(-\theta) = -\sin\theta$.
(c)(ii) Substitute $m = n$ first; $\sin mx\cos mx = \frac{1}{2}\sin 2mx$ via the double-angle formula.
(d) Show the integrand is an odd function. The integral of any odd function over a symmetric interval $[-a, a]$ is zero.

Solution 2.16

(a) Expanding the compound-angle formulae:

$$\sin(A+B) = \sin A\cos B + \cos A\sin B, \qquad \sin(A-B) = \sin A\cos B - \cos A\sin B.$$

Adding: $\sin(A+B) + \sin(A-B) = 2\sin A\cos B$. □

(b)(i) Set $A = 3x$, $B = 2x$: $2\sin 3x\cos 2x = \sin 5x + \sin x$.

$$\int_0^{\pi/2} (\sin 5x + \sin x)\,dx = \left[-\tfrac{1}{5}\cos 5x - \cos x\right]_0^{\pi/2} = (0-0) - (-\tfrac{1}{5} - 1) = \frac{6}{5}.$$

(b)(ii) Set $A = 2x$, $B = 4x$: $\sin 2x\cos 4x = \frac{1}{2}[\sin 6x + \sin(-2x)] = \frac{1}{2}(\sin 6x - \sin 2x)$.

$$\int_0^{\pi} \tfrac{1}{2}(\sin 6x - \sin 2x)\,dx = \tfrac{1}{2}\left[-\tfrac{1}{6}\cos 6x + \tfrac{1}{2}\cos 2x\right]_0^{\pi} = \tfrac{1}{2}\left[(-\tfrac{1}{6} + \tfrac{1}{2}) - (-\tfrac{1}{6} + \tfrac{1}{2})\right] = 0.$$

(c)(i) $m \neq n$:

$$\int \sin mx\cos nx\,dx = \tfrac{1}{2}\int \left[\sin(m+n)x + \sin(m-n)x\right]dx = -\frac{\cos(m+n)x}{2(m+n)} - \frac{\cos(m-n)x}{2(m-n)} + C.$$

(c)(ii) $m = n$: $\sin mx\cos mx = \frac{1}{2}\sin 2mx$, so

$$\int \sin mx\cos mx\,dx = \frac{1}{2}\int \sin 2mx\,dx = -\frac{\cos 2mx}{4m} + C.$$

(d) Let $f(x) = \sin mx\cos nx$. Then

$$f(-x) = \sin(-mx)\cos(-nx) = (-\sin mx)(\cos nx) = -f(x),$$

so f is odd. The integral of any odd function over a symmetric interval $[-\pi, \pi]$ is zero. Alternatively, using (c)(i): $\cos(k\pi) = \cos(-k\pi)$ for any integer k, so the upper and lower limit evaluations are equal and their difference is zero.

Takeaways 2.16

- The identity $2\sin A\cos B = \sin(A+B) + \sin(A-B)$ converts products into sums that are straightforward to integrate.
- When $m = n$, the double-angle shortcut $\sin(mx)\cos(mx) = \frac{1}{2}\sin(2mx)$ avoids the product-to-sum formula altogether.
- Recognising an integrand as an odd function over a symmetric interval immediately gives a zero result—a cornerstone of Fourier analysis.

Problem 2.17: Establishing the Lower Bound of $\sin x$

Let $f(x) = \sin x - x + \dfrac{x^3}{6}$ for $x \geq 0$.

(i) Find $f'(x)$, $f''(x)$, $f'''(x)$, and $f^{(4)}(x)$.

(ii) By considering the values of these derivatives at $x = 0$ and their signs for $x > 0$, prove that

$$\sin x \geq x - \frac{x^3}{6}$$

for all $x \geq 0$.

Hint 2.15

Chain of Logic: If $g'(x) \geq 0$ for all $x \geq 0$ and $g(0) = 0$, then $g(x) \geq 0$ for all $x \geq 0$. Start from $f^{(4)}(x)$ and work downwards: show $f'''(x) \geq 0$, then $f''(x) \geq 0$, then $f'(x) \geq 0$, and finally $f(x) \geq 0$. Each step uses only the sign of the derivative above it and the initial condition at $x = 0$.

Solution 2.17

(i) Derivatives.

$$f'(x) = \cos x - 1 + \tfrac{x^2}{2}, \quad f''(x) = -\sin x + x, \quad f'''(x) = -\cos x + 1, \quad f^{(4)}(x) = \sin x.$$

(ii) Proof by successive monotonicity.

Step 1. For $x \in [0, \pi]$, $f^{(4)}(x) = \sin x \geq 0$.

Step 2. Since $f^{(4)} \geq 0$, f''' is non-decreasing on $[0, \infty)$. Since $f'''(0) = -\cos 0 + 1 = 0$, we have $f'''(x) \geq 0$ for all $x \geq 0$.

Step 3. Since $f''' \geq 0$, f'' is non-decreasing. Since $f''(0) = -\sin 0 + 0 = 0$, we have $f''(x) \geq 0$ for all $x \geq 0$.

Step 4. Since $f'' \geq 0$, f' is non-decreasing. Since $f'(0) = \cos 0 - 1 + 0 = 0$, we have $f'(x) \geq 0$ for all $x \geq 0$.

Step 5. Since $f' \geq 0$, f is non-decreasing. Since $f(0) = \sin 0 - 0 + 0 = 0$, we have $f(x) \geq 0$ for all $x \geq 0$.

Therefore $\sin x - x + \dfrac{x^3}{6} \geq 0$, which rearranges to

$$\sin x \geq x - \frac{x^3}{6}. \qquad \square$$

Takeaways 2.17

- This "successive monotonicity" technique proves polynomial bounds on transcendental functions without invoking the Taylor series formula directly.
- The initial conditions $f(0) = f'(0) = f''(0) = f'''(0) = 0$ allow each inequality to propagate upward from the origin.
- Along the way, Step 3 proves the classical inequality $\sin x \leq x$ (since $f''(x) \geq 0$ means $x - \sin x \geq 0$).

Problem 2.18: Polynomial Bounds and the Cosine Series

The Maclaurin series for $\cos x$ is

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \text{ Do NOT prove this}$$

(i) It is known geometrically that $\sin t \le t$ for $t \ge 0$. Integrate this inequality over $[0, x]$ (with $x > 0$) to show that $\cos x \ge 1 - \dfrac{x^2}{2!}$.

(ii) Rearrange the inequality from (i) to the form $1 - \dfrac{t^2}{2!} \le \cos t$. Integrate this inequality twice over $[0, x]$ to show that

$$\cos x \le 1 - \frac{x^2}{2!} + \frac{x^4}{4!}.$$

(iii) By continuing this process of repeated integration, the pattern alternates between upper and lower bounds. Write down the compound inequality bounding $\cos x$ between its degree-$(4n+2)$ lower bound and degree-$4n$ upper bound, for any integer $n \ge 0$.

(iv) Use the concept of even functions to explain briefly why the series converges to $\cos x$ for *all* real x, not just $x \ge 0$.

Hint 2.16

(i) Recall $\int_0^x \sin t \, dt = 1 - \cos x$. Be careful with the signs.
(ii) First integrate $1 - t^2/2 \le \cos t$ to obtain a bound on $\sin x$, then integrate the sine bound to return to a bound on $\cos x$.
(iii) When the highest-power term is x^{4n+2} the sign is negative (lower bound); when it is x^{4n} the sign is positive (upper bound).
(iv) What does substituting $-x$ do to all even powers x^{2k}?

Solution 2.18

(i) Integrating $\sin t \le t$ from 0 to x:

$$\int_0^x \sin t\,dt \le \int_0^x t\,dt \implies [-\cos t]_0^x \le \tfrac{x^2}{2} \implies 1-\cos x \le \tfrac{x^2}{2!},$$

so $\cos x \ge 1 - \dfrac{x^2}{2!}$.

(ii) From (i), $1 - \dfrac{t^2}{2!} \le \cos t$. Integrate from 0 to x:

$$x - \frac{x^3}{3!} \le \sin x.$$

Now integrate $t - \dfrac{t^3}{3!} \le \sin t$ from 0 to x:

$$\frac{x^2}{2!} - \frac{x^4}{4!} \le [-\cos t]_0^x = 1 - \cos x,$$

so $\cos x \le 1 - \dfrac{x^2}{2!} + \dfrac{x^4}{4!}$.

(iii) Continuing the pattern:

$$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots - \frac{x^{4n+2}}{(4n+2)!} \;\le\; \cos x \;\le\; 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + \frac{x^{4n}}{(4n)!}.$$

As $n \to \infty$, factorials grow faster than any exponential, so both bounds converge to $\cos x$ for all $x > 0$.

(iv) Since $\cos x$ is an even function and every term x^{2k} satisfies $(-x)^{2k} = x^{2k}$, substituting $-x$ leaves the series unchanged. The bounds established for $x > 0$ therefore hold identically for $x < 0$.

Takeaways 2.18

- Repeated integration is a rigorous way to derive Taylor polynomial bounds from a single geometric inequality ($\sin t \le t$), without invoking the Taylor formula itself.
- The factorials in the denominators arise naturally from repeated integration of powers: $\int t^n\,dt = t^{n+1}/(n+1)$.
- The even-function argument extends results from the non-negative half-axis to the full real line without any extra calculation.

Problem 2.19: The Laplace Transform of Cosine

Let $a > 0$. Consider the improper integral $I = \displaystyle\int_0^\infty e^{-ax}\cos x\,dx$. Using integration by parts twice, show that $I = \dfrac{a}{a^2+1}$.

Hint 2.17

Let $u = \cos x$, $dv = e^{-ax}dx$. After two applications of parts, the original integral I reappears on the right-hand side; collect it algebraically. Recall $e^{-ax} \to 0$ as $x \to \infty$ (since $a > 0$).

Solution 2.19

Let $u = \cos x$, $dv = e^{-ax}dx$, so $du = -\sin x\,dx$, $v = -\frac{1}{a}e^{-ax}$.

$$I = \left[-\frac{e^{-ax}\cos x}{a}\right]_0^\infty - \int_0^\infty \frac{e^{-ax}\sin x}{a}\,dx = \frac{1}{a} - \frac{1}{a}\int_0^\infty e^{-ax}\sin x\,dx.$$

Apply parts to $\int e^{-ax}\sin x\,dx$ with $u = \sin x$, $dv = e^{-ax}dx$:

$$\int_0^\infty e^{-ax}\sin x\,dx = \left[-\frac{e^{-ax}\sin x}{a}\right]_0^\infty + \frac{1}{a}\int_0^\infty e^{-ax}\cos x\,dx = 0 + \frac{I}{a}.$$

Substituting back:

$$I = \frac{1}{a} - \frac{1}{a}\cdot\frac{I}{a} = \frac{1}{a} - \frac{I}{a^2}.$$

Solving: $I\left(1 + \frac{1}{a^2}\right) = \frac{1}{a}$, so

$$\mathbf{I = \frac{a}{a^2+1}}. \qquad \square$$

Takeaways 2.19

- The "looping" integration by parts technique works whenever the integrand reproduces itself after an even number of applications—as it does for $e^{ax}\sin(bx)$ and $e^{ax}\cos(bx)$.
- This integral is the **Laplace transform** of $\cos x$, written $\mathcal{L}\{\cos x\}(a) = \dfrac{a}{a^2+1}$, a fundamental tool for solving differential equations at university level.
- **Euler's formula shortcut (Extension 2 / university):** Consider the complex integral $Z = \int_0^\infty e^{-(a-i)x}\,dx = \dfrac{1}{a-i}$. Rationalizing gives $Z = \dfrac{a+i}{a^2+1}$. By Euler's formula $e^{ix} = \cos x + i\sin x$, the real part of Z equals I, yielding $I = \dfrac{a}{a^2+1}$ immediately—and the imaginary part gives $\int_0^\infty e^{-ax}\sin x\,dx = \dfrac{1}{a^2+1}$ for free.

Problem 2.20: De Moivre's Theorem and $\cos 5\theta$

(i) Using De Moivre's theorem, show that

$$\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta.$$

(ii) Hence find all real roots of the equation

$$16x^5 - 20x^3 + 5x = 1$$

in exact form, and express them as cosines of rational multiples of π.

Solution 2.20

(i) Write $c = \cos\theta$, $s = \sin\theta$. By De Moivre's theorem and the binomial theorem:

$$(c + is)^5 = c^5 + 5ic^4s - 10c^3s^2 - 10ic^2s^3 + 5cs^4 + is^5.$$

Taking the real part and substituting $s^2 = 1 - c^2$:

$$\cos 5\theta = c^5 - 10c^3s^2 + 5cs^4 = c^5 - 10c^3(1 - c^2) + 5c(1 - c^2)^2 = 16c^5 - 20c^3 + 5c. \quad \square$$

(ii) Substitute $x = \cos\theta$: the equation becomes $\cos 5\theta = 1$, so $5\theta = 2k\pi$, giving $\theta = \frac{2k\pi}{5}$. For $\theta \in [0, 2\pi)$ the five values are $k = 0, 1, 2, 3, 4$; using $\cos(2\pi - \alpha) = \cos\alpha$:

$$\cos\tfrac{6\pi}{5} = \cos\tfrac{4\pi}{5}, \qquad \cos\tfrac{8\pi}{5} = \cos\tfrac{2\pi}{5}.$$

Hence the three distinct real roots are:

$$\boxed{x = 1, \quad x = \cos\tfrac{2\pi}{5} = \tfrac{\sqrt{5}-1}{4}, \quad x = \cos\tfrac{4\pi}{5} = -\tfrac{\sqrt{5}+1}{4}.}$$

($x = \cos(2\pi/5)$ and $x = \cos(4\pi/5)$ are each double roots of the degree-5 polynomial.)

Takeaways 2.20

- De Moivre's theorem $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$ converts $\cos n\theta$ into a polynomial in $\cos\theta$ by taking the real part and using $\sin^2\theta = 1 - \cos^2\theta$.
- The fifth-degree Chebyshev polynomial $T_5(x) = 16x^5 - 20x^3 + 5x$ satisfies $T_5(\cos\theta) = \cos 5\theta$; this is the general pattern behind the triple-angle formulae in the appendix.
- Solving $\cos n\theta = c$ geometrically by listing all n values of θ in $[0, 2\pi)$ automatically produces all roots, including repeated ones, of the corresponding polynomial equation.

3 Part 2: Practice and Challenge Bank

3.1 Warm-Up Drills

Problem 3.1: Principal Value of an Inverse Sine

Calculate

$$\sin^{-1}\left(\sin\frac{5}{2}\right),$$

rounding your answer to 3 decimal places. Angles are measured in radians.

Hint 3.1

The range of $\sin^{-1} x$ is $[-\pi/2, \pi/2]$. The angle $\frac{5}{2}$ radians is not in that interval, but $\pi - \frac{5}{2}$ is.

Solution 3.1

Since $\frac{5}{2} \in (\pi/2, \pi)$,

$$\sin\frac{5}{2} = \sin\left(\pi - \frac{5}{2}\right).$$

The angle $\pi - \frac{5}{2}$ lies in the principal range of arcsin, because $0 < \pi - \frac{5}{2} < \pi/2$. Hence

$$\sin^{-1}\left(\sin\frac{5}{2}\right) = \pi - \frac{5}{2} \approx 0.642.$$

Takeaways 3.1

- $\sin^{-1}(\sin x)$ does not always equal x.
- Always return the principal value in $[-\pi/2, \pi/2]$.
- Supplementary angles have equal sine values.

Problem 3.2: A Sine-Cosine Equation

Solve

$$\sin(3x) = \cos\left(2x - \frac{\pi}{4}\right)$$

for $0 \le x \le 2\pi$.

Hint 3.2

Convert cosine to sine using $\cos u = \sin(\pi/2 - u)$, then solve $\sin\alpha = \sin\beta$.

Solution 3.2

$$\cos\left(2x - \frac{\pi}{4}\right) = \sin\left(\frac{\pi}{2} - 2x + \frac{\pi}{4}\right) = \sin\left(\frac{3\pi}{4} - 2x\right).$$

Thus

$$\sin(3x) = \sin\left(\frac{3\pi}{4} - 2x\right).$$

For $\sin A = \sin B$,

$$A = B + 2k\pi \quad \text{or} \quad A = \pi - B + 2k\pi.$$

Hence

$$3x = \frac{3\pi}{4} - 2x + 2k\pi \quad \Longrightarrow \quad x = \frac{3\pi}{20} + \frac{2k\pi}{5},$$

or

$$3x = \pi - \left(\frac{3\pi}{4} - 2x\right) + 2k\pi \quad \Longrightarrow \quad x = \frac{\pi}{4} + 2k\pi.$$

Restricting to $0 \le x \le 2\pi$ gives

$$x = \frac{3\pi}{20}, \frac{11\pi}{20}, \frac{19\pi}{20}, \frac{27\pi}{20}, \frac{35\pi}{20}, \frac{\pi}{4}.$$

Takeaways 3.2

- Convert mixed sine-cosine equations into one trigonometric function.
- The equation $\sin A = \sin B$ has two families of solutions.
- Restrict the final answer to the requested interval.

Problem 3.3: Secant-Tangent Convergence

Let $0 \le A \le \pi$.

(i) Prove that

$$\frac{1}{1-\tan A} - \frac{1}{1+\tan A} + \frac{2\tan A}{\sec^2 A - 2\tan^2 A} = 2\tan 2A,$$

where both sides are defined.

(ii) Hence find all values of A such that

$$\frac{1}{1-\tan A} - \frac{1}{1+\tan A} = 1.$$

Hint 3.3

Combine the first two fractions. For the third term, use $\sec^2 A = 1 + \tan^2 A$.

Solution 3.3

The first two terms combine to

$$\frac{(1+\tan A)-(1-\tan A)}{(1-\tan A)(1+\tan A)} = \frac{2\tan A}{1-\tan^2 A} = \tan 2A.$$

Also,

$$\sec^2 A - 2\tan^2 A = 1 + \tan^2 A - 2\tan^2 A = 1 - \tan^2 A,$$

so

$$\frac{2\tan A}{\sec^2 A - 2\tan^2 A} = \frac{2\tan A}{1-\tan^2 A} = \tan 2A.$$

Therefore the full expression is $2\tan 2A$.
For part (ii), the left side is $\tan 2A$, so

$$\tan 2A = 1.$$

Since $0 \le A \le \pi$, we have $0 \le 2A \le 2\pi$. Thus

$$2A = \frac{\pi}{4}, \frac{5\pi}{4},$$

and

$$A = \frac{\pi}{8}, \frac{5\pi}{8}.$$

Takeaways 3.3

- Expressions involving $1 - \tan^2 A$ often point to $\tan 2A$.
- Reciprocal identities can reveal hidden double-angle forms.
- When solving $\tan 2A$, double the interval before listing solutions.

Problem 3.4: An Inverse Sine and Cosine Identity

Prove or disprove the statement:

$$\arcsin x + \arccos x = \frac{\pi}{2}$$

for any real number x for which the left-hand side is defined.

Hint 3.4

First check the domain. Both $\arcsin x$ and $\arccos x$ require $-1 \leq x \leq 1$.

Solution 3.4

The left-hand side is defined exactly when both inverse trigonometric functions are defined, so we must have

$$-1 \leq x \leq 1.$$

For such an x, let

$$\theta = \arcsin x.$$

Then $\theta \in [-\pi/2, \pi/2]$ and $\sin\theta = x$. Hence

$$\cos\left(\frac{\pi}{2} - \theta\right) = \sin\theta = x.$$

Since $\frac{\pi}{2} - \theta \in [0, \pi]$, this angle lies in the principal range of arccos. Therefore

$$\arccos x = \frac{\pi}{2} - \theta.$$

Thus

$$\arcsin x + \arccos x = \theta + \left(\frac{\pi}{2} - \theta\right) = \frac{\pi}{2}.$$

Takeaways 3.4

- Always check the domain before deciding whether an identity is true.
- The identity $\arcsin x + \arccos x = \pi/2$ is valid exactly for $x \in [-1, 1]$.
- Principal ranges are part of the definition of inverse trigonometric functions.

Problem 3.5: Counting Roots with a Sum-to-Product Identity

How many distinct solutions are there to

$$\cos 4x + \cos 2x = 0$$

in the interval $0 \le x \le 2\pi$?

A. 4 B. 6 C. 8 D. 12

Hint 3.5

Use $\cos P + \cos Q = 2\cos\frac{P+Q}{2}\cos\frac{P-Q}{2}$, then check whether the solution sets overlap.

Solution 3.5

Using the sum-to-product identity,

$$\cos 4x + \cos 2x = 2\cos 3x \cos x.$$

Thus

$$2\cos 3x \cos x = 0,$$

so either $\cos x = 0$ or $\cos 3x = 0$.
For $\cos x = 0$ on $[0, 2\pi]$,

$$x = \frac{\pi}{2}, \frac{3\pi}{2}.$$

For $\cos 3x = 0$,

$$3x = \frac{\pi}{2} + k\pi.$$

Since $0 \le 3x \le 6\pi$, we have

$$3x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \frac{9\pi}{2}, \frac{11\pi}{2},$$

giving six values of x.
The two roots from $\cos x = 0$ are already included in the roots from $\cos 3x = 0$, so the total number of distinct solutions is

$$\boxed{6}.$$

The correct answer is B.

Takeaways 3.5

- Sum-to-product identities convert some equations into factored form.
- When solving $\cos(nx) = 0$, scale the argument interval carefully.
- Count distinct roots; do not double-count roots shared by two factors.

3.2 Stretch Problems

Problem 3.6: Telescoping Cosines and Periodic Constants

(i) Use $2\sin A\cos B = \sin(A+B) + \sin(A-B)$ to show that

$$2\sin x(\cos 2x + \cos 4x + \cos 6x) = \sin 7x - \sin x.$$

(ii) By substituting $x = \pi/7$, deduce that

$$\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7} = -\frac{1}{2}.$$

(iii) Use supplementary angles to show that

$$\cos\frac{6\pi}{7} = -\cos\frac{\pi}{7}, \quad \cos\frac{4\pi}{7} = -\cos\frac{3\pi}{7}, \quad \cos\frac{2\pi}{7} = -\cos\frac{5\pi}{7}.$$

(iv) Hence evaluate

$$\cos\frac{\pi}{7} + \cos\frac{3\pi}{7} + \cos\frac{5\pi}{7}.$$

Hint 3.6

Distribute $2\sin x$ and watch terms such as $\sin 3x$ and $\sin 5x$ cancel. Then use $\sin\pi = 0$ and $\sin(\pi/7) \neq 0$.

Solution 3.6

$$\begin{aligned}2\sin x(\cos 2x + \cos 4x + \cos 6x) &= (\sin 3x - \sin x) + (\sin 5x - \sin 3x)\\ &\quad + (\sin 7x - \sin 5x)\\ &= \sin 7x - \sin x.\end{aligned}$$

Putting $x = \pi/7$ gives

$$2\sin\frac{\pi}{7}\left(\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7}\right) = -\sin\frac{\pi}{7}.$$

Since $\sin(\pi/7) \neq 0$,

$$\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7} = -\frac{1}{2}.$$

Using $\cos(\pi - \theta) = -\cos\theta$,

$$\cos\frac{6\pi}{7} = -\cos\frac{\pi}{7}, \quad \cos\frac{4\pi}{7} = -\cos\frac{3\pi}{7}, \quad \cos\frac{2\pi}{7} = -\cos\frac{5\pi}{7}.$$

Therefore

$$-\left(\cos\frac{\pi}{7} + \cos\frac{3\pi}{7} + \cos\frac{5\pi}{7}\right) = -\frac{1}{2},$$

so the required sum is $\frac{1}{2}$.

Takeaways 3.6

- Multiplying by a carefully chosen sine can turn a cosine sum into a telescoping expression.
- Regular polygon angle sums often hide cancellation.
- Supplementary angles convert even seventh-angle cosines into odd seventh-angle cosines.

Problem 3.7: The Summation of Even Sine Multiples

Let
$$S_n = \sin 2x + \sin 4x + \sin 6x + \cdots + \sin(2nx).$$

(i) Use $2\sin A \sin B = \cos(A - B) - \cos(A + B)$ to show that
$$2\sin(2kx)\sin x = \cos(2k - 1)x - \cos(2k + 1)x.$$

(ii) Hence prove that
$$\sin 2x + \sin 4x + \cdots + \sin(2nx) = \frac{\sin(nx)\sin((n + 1)x)}{\sin x},$$
for $\sin x \neq 0$.

Hint 3.7

Write out the first few terms of $2\sin x\, S_n$. Only the first and last cosine terms survive.

Solution 3.7

$$2\sin(2kx)\sin x = \cos(2kx - x) - \cos(2kx + x) = \cos(2k - 1)x - \cos(2k + 1)x.$$

Thus
$$\begin{aligned} 2\sin x\, S_n &= \sum_{k=1}^{n}\{\cos(2k - 1)x - \cos(2k + 1)x\} \\ &= (\cos x - \cos 3x) + (\cos 3x - \cos 5x) + \cdots \\ &\quad + (\cos(2n - 1)x - \cos(2n + 1)x) \\ &= \cos x - \cos(2n + 1)x. \end{aligned}$$

Now
$$\cos x - \cos(2n + 1)x = -2\sin((n + 1)x)\sin(-nx) = 2\sin(nx)\sin((n + 1)x).$$

Dividing by $2\sin x$ gives
$$S_n = \frac{\sin(nx)\sin((n + 1)x)}{\sin x}.$$

Takeaways 3.7

- Telescoping trigonometric sums often begin by multiplying by $2\sin$ of half the step size.
- The remaining endpoint terms are then simplified with a sum-to-product identity.
- Always state any division restriction, here $\sin x \neq 0$.

Problem 3.8: Root Analysis of High-Frequency Oscillations

Consider

$$f(x) = \cos(2025x) - \cos(2026x).$$

Determine the total number of roots of $f(x) = 0$ in the closed interval $[0, \pi]$.

Hint 3.8

Use $\cos P - \cos Q = -2\sin\frac{P+Q}{2}\sin\frac{P-Q}{2}$. Count distinct solutions from the two sine factors.

Solution 3.8

$$\begin{aligned} f(x) &= -2\sin\left(\frac{2025x + 2026x}{2}\right)\sin\left(\frac{2025x - 2026x}{2}\right) \\ &= 2\sin\left(\frac{4051x}{2}\right)\sin\left(\frac{x}{2}\right). \end{aligned}$$

Therefore $f(x) = 0$ when

$$\sin\left(\frac{x}{2}\right) = 0 \quad \text{or} \quad \sin\left(\frac{4051x}{2}\right) = 0.$$

On $[0, \pi]$, the first factor gives only $x = 0$.
The second factor gives

$$\frac{4051x}{2} = k\pi \quad \Longrightarrow \quad x = \frac{2k\pi}{4051}.$$

The condition $0 \leq x \leq \pi$ becomes

$$0 \leq k \leq \frac{4051}{2} = 2025.5.$$

Thus $k = 0, 1, 2, \ldots, 2025$, giving 2026 distinct roots. The root $x = 0$ is already included.

$$\boxed{2026}$$

Takeaways 3.8

- A high-frequency difference of cosines is much easier to count in product form.
- Closed intervals require careful endpoint counting.
- Count the parameter values first, then check for duplicate roots.

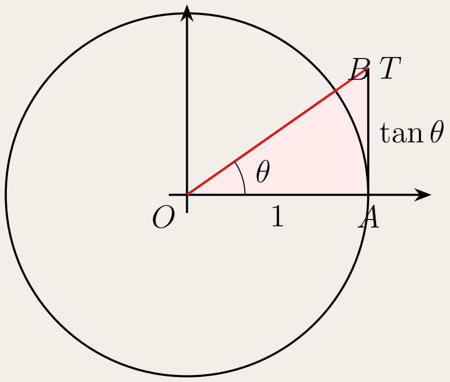
Problem 3.9: Bounds and Inequalities of Trigonometric Ratios

(i) Using areas in a unit circle, or otherwise, prove that

$$\tan\theta > \theta, \qquad 0 < \theta < \frac{\pi}{2}.$$

(ii) Hence, or otherwise, show that

$$\frac{\tan\theta}{\theta} > \frac{\theta}{\sin\theta}, \qquad 0 < \theta < \frac{\pi}{2}.$$



Hint 3.9

For part (i), compare the sector OAB with the tangent triangle OAT. For part (ii), one concise path is to prove $\sin\theta\tan\theta - \theta^2 > 0$ by calculus.

Solution 3.9

In the unit circle diagram, the sector OAB has area $\theta/2$, while the tangent triangle OAT has area $\tan\theta/2$. Since the sector lies strictly inside the tangent triangle for $0 < \theta < \pi/2$,

$$\frac{\theta}{2} < \frac{\tan\theta}{2},$$

so $\tan\theta > \theta$.

For the second inequality, it is equivalent to prove

$$\sin\theta\tan\theta > \theta^2.$$

Define

$$F(\theta) = \sin\theta\tan\theta - \theta^2.$$

Then $F(0) = 0$ in the limiting sense, and

$$F'(\theta) = \sin\theta + \sin\theta\sec^2\theta - 2\theta.$$

Differentiating again,

$$F''(\theta) = \cos\theta + \sec\theta + 2\sin^2\theta\sec^3\theta - 2.$$

Since $\cos\theta + \sec\theta \geq 2$ for $0 < \theta < \pi/2$, and the final term is positive,

$$F''(\theta) > 0.$$

Thus $F'(\theta) > F'(0) = 0$, and hence $F(\theta) > F(0) = 0$ for $0 < \theta < \pi/2$. Therefore

$$\sin\theta\tan\theta > \theta^2,$$

which gives

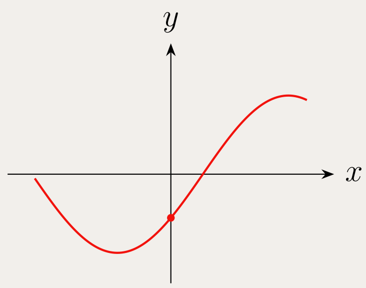
$$\frac{\tan\theta}{\theta} > \frac{\theta}{\sin\theta}.$$

Takeaways 3.9

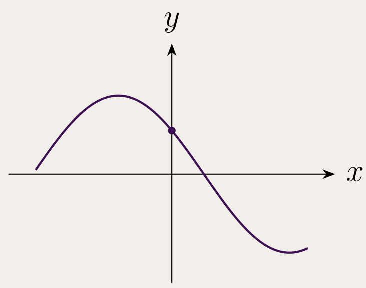
- Area comparisons give elegant proofs of basic trigonometric inequalities.
- Some stronger inequalities need a more precise tool, such as a derivative sign argument.
- Before proving a ratio inequality, cross-multiply carefully and check all quantities are positive.

Problem 3.10: Graphical Analysis of Harmonic Functions

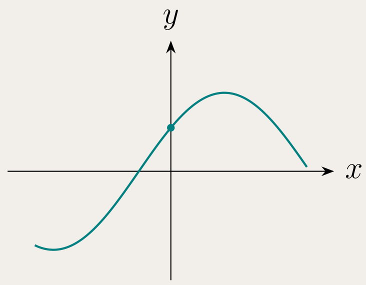
Let
$$f(x) = a\sin x - b\cos x,$$
where $a > 0$ and $b > 0$. Which of the following curves could represent f near the origin?



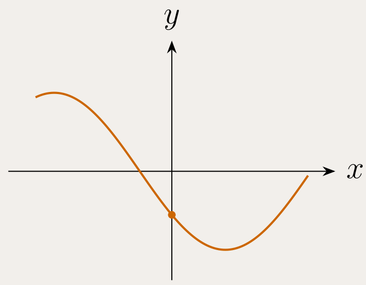
Graph A



Graph B



Graph C



Graph D

Hint 3.10

Use the y-intercept and the derivative at $x = 0$. These two quick tests often identify the correct graph before any full phase-shift calculation is needed.

Solution 3.10

At the origin,
$$f(0) = a\sin 0 - b\cos 0 = -b.$$
Since $b > 0$, the graph must have a negative y-intercept.
Next,
$$f'(x) = a\cos x + b\sin x,$$
so
$$f'(0) = a > 0.$$
Thus the curve must be increasing at $x = 0$.
Among the four sketches, the only graph with a negative y-intercept and a positive gradient at $x = 0$ is Graph A.

$$\boxed{\text{Graph A}}$$

Takeaways 3.10

- The intercept $f(0)$ is a fast way to eliminate graph options.
- The sign of $f'(0)$ tells whether the curve is rising or falling at the y-axis.
- Auxiliary-angle form confirms that a sine-cosine combination is still a shifted sinusoid.

Problem 3.11: Simplifying an Inverse Trigonometric Difference

Simplify
$$\arcsin(\cos x) - \arccos(\sin x),$$
where x is a real number.

Hint 3.11

Because inverse trigonometric functions return principal values, the answer is piecewise. First solve it on $0 \le x \le 2\pi$, then extend periodically.

Solution 3.11

Both $\arcsin(\cos x)$ and $\arccos(\sin x)$ are 2π-periodic, so it is enough to work on $0 \le x \le 2\pi$. On this interval,
$$\arcsin(\cos x) - \arccos(\sin x) = \begin{cases} 0, & 0 \le x \le \dfrac{\pi}{2}, \\ \pi - 2x, & \dfrac{\pi}{2} \le x \le \pi, \\ -\pi, & \pi \le x \le \dfrac{3\pi}{2}, \\ 2x - 4\pi, & \dfrac{3\pi}{2} \le x \le 2\pi. \end{cases}$$
For all real x, reduce x modulo 2π and apply this piecewise formula.

Takeaways 3.11

- Expressions like $\arcsin(\sin x)$ and $\arccos(\cos x)$ are governed by principal ranges.
- A single simple-looking inverse-trigonometric expression can require a piecewise answer.
- Periodicity lets us solve on one full period and extend to all real x.

Problem 3.12: A Numerical Trigonometric Equation

Solve for $0 \le x \le 2\pi$:
$$4\sin^3 x + 8\sin^2 x - 19\sin x + 7 = 0.$$

Hint 3.12

Put $u = \sin x$. Notice that the resulting cubic has the root $u = 1$.

Solution 3.12

Put $u = \sin x$. The equation becomes

$$4u^3 + 8u^2 - 19u + 7 = 0.$$

Since $u = 1$ is a root,

$$4u^3 + 8u^2 - 19u + 7 = (u - 1)(4u^2 + 12u - 7).$$

Thus

$$u = 1, \qquad 4u^2 + 12u - 7 = 0.$$

The quadratic gives

$$u = \frac{-12 \pm 16}{8}, \quad \text{so} \quad u = \frac{1}{2} \quad \text{or} \quad u = -\frac{7}{2}.$$

Because $u = \sin x$, the value $u = -\frac{7}{2}$ is impossible. Hence

$$\sin x = 1 \quad \text{or} \quad \sin x = \frac{1}{2}.$$

For $0 \le x \le 2\pi$, the solutions are

$$\boxed{x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}}.$$

Takeaways 3.12

- If an equation is a polynomial in $\sin x$, try the substitution $u = \sin x$.
- After solving for u, reject any values outside the range $-1 \le u \le 1$.
- Convert the remaining sine values back to all angles in the required interval.

3.3 Challenge Corner

Problem 3.13: The Iterated Polynomial

Define $f(x) = 2x^2 - 1$ on the interval $-1 \le x \le 1$. The *nth iterate* $f^n(x)$ means f applied n times in succession; for example, $f^2(x) = f(f(x))$ and $f^3(x) = f(f(f(x)))$.

(i) Using $x = \cos\theta$, where $0 \le \theta \le \pi$, show that $f(x) = \cos 2\theta$.

(ii) Prove by induction that, for every integer $n \ge 1$,

$$f^n(x) = \cos(2^n\theta).$$

(iii) Hence determine the exact number of distinct real solutions of

$$f^3(x) = x.$$

(iv) Deduce the number of distinct real roots of

$$f^n(x) = 0$$

in terms of n.

Hint 3.13

The identity $2\cos^2\theta - 1 = \cos 2\theta$ is doing all the work. For $\cos A = \cos B$, use $A = 2k\pi \pm B$. The restriction $0 \le \theta \le \pi$ makes $x = \cos\theta$ one-to-one.

Solution 3.13

If $x = \cos\theta$, then

$$f(x) = 2\cos^2\theta - 1 = \cos 2\theta.$$

We prove the iterate formula by induction. For $n = 1$ it is exactly the result above. Suppose

$$f^k(x) = \cos(2^k\theta).$$

Then

$$f^{k+1}(x) = f(f^k(x)) = f(\cos(2^k\theta)) = 2\cos^2(2^k\theta) - 1 = \cos(2^{k+1}\theta).$$

Hence, by induction,

$$f^n(x) = \cos(2^n\theta)$$

for all $n \geq 1$.

For $f^3(x) = x$, we solve

$$\cos 8\theta = \cos\theta, \qquad 0 \leq \theta \leq \pi.$$

Thus

$$8\theta = 2k\pi + \theta \quad \text{or} \quad 8\theta = 2k\pi - \theta.$$

The first family gives

$$\theta = \frac{2k\pi}{7},$$

so $k = 0, 1, 2, 3$, giving 4 values in $[0, \pi]$. The second gives

$$\theta = \frac{2k\pi}{9},$$

so $k = 0, 1, 2, 3, 4$, giving 5 values, but $\theta = 0$ has already been counted. Therefore there are $4 + 4 = 8$ distinct values of θ. Since $\cos\theta$ is one-to-one on $[0, \pi]$, there are exactly

$$\boxed{8}$$

distinct real solutions for x.

Finally, $f^n(x) = 0$ becomes

$$\cos(2^n\theta) = 0.$$

So

$$2^n\theta = \frac{(2k+1)\pi}{2}, \qquad \theta = \frac{(2k+1)\pi}{2^{n+1}}.$$

We need $0 \leq \theta \leq \pi$, which is equivalent to

$$0 \leq 2k + 1 \leq 2^{n+1}.$$

There are exactly 2^n positive odd integers not exceeding 2^{n+1}. Hence $f^n(x) = 0$ has

$$\boxed{2^n}$$

distinct real roots.

Takeaways 3.13

- Some high-degree polynomial equations become simple after the substitution $x = \cos\theta$.
- The restriction $0 \leq \theta \leq \pi$ prevents double-counting values of x.
- Root-counting often matters more than explicitly writing every root.

Problem 3.14: A Cubic Trigonometric Equation

The **triple-angle formula** states that $\cos 3x = 4\cos^3 x - 3\cos x$.
Use the triple-angle formula to solve

$$4\cos^3 x - 3\cos x = \frac{1}{2}$$

for $0 \leq x \leq 2\pi$.

Hint 3.14

Recognise $4\cos^3 x - 3\cos x$ as $\cos 3x$. Then solve a standard cosine equation, remembering that $3x$ ranges over a longer interval.

Solution 3.14

Using

$$\cos 3x = 4\cos^3 x - 3\cos x,$$

the equation becomes

$$\cos 3x = \frac{1}{2}.$$

Since $0 \leq x \leq 2\pi$, we have $0 \leq 3x \leq 6\pi$. Therefore

$$3x = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}, \frac{13\pi}{3}, \frac{17\pi}{3}.$$

Dividing by 3 gives

$$\boxed{x = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}, \frac{11\pi}{9}, \frac{13\pi}{9}, \frac{17\pi}{9}}.$$

Takeaways 3.14

- The cubic expression $4u^3 - 3u$ is a signal to look for $\cos 3x$ or the Chebyshev polynomial $T_3(u)$.
- Triple-angle formulae can turn cubic-looking equations into simple trigonometric equations.
- When the argument is $3x$, solve over $0 \leq 3x \leq 6\pi$ before dividing by 3.

Problem 3.15: Jensen's Inequality Applied to Sine

Jensen's Inequality (stated without proof): For any concave function f and three numbers a_1, a_2, a_3,

$$f(a_1) + f(a_2) + f(a_3) \leq 3\, f\left(\frac{a_1 + a_2 + a_3}{3}\right).$$

Since sin is concave on $(0, \pi)$, this inequality applies to sin when a_1, a_2, a_3 are angles of a triangle.
Let A, B, C be the interior angles of a triangle. Prove that

$$\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2},$$

with equality if and only if the triangle is equilateral.

Hint 3.15

Step 1. Show that $f(x) = \sin x$ is concave on $(0, \pi)$ by computing $f''(x)$.
Step 2. Apply Jensen's inequality: for a concave function f and weights summing to 1,

$$f\left(\frac{x_1 + x_2 + x_3}{3}\right) \geq \frac{f(x_1) + f(x_2) + f(x_3)}{3}.$$

Use $A + B + C = \pi$.

Solution 3.15

Step 1: Concavity of $\sin$ on $(0, \pi)$.
Let $f(x) = \sin x$. Then $f''(x) = -\sin x$. For $x \in (0, \pi)$, $\sin x > 0$, so $f''(x) < 0$ throughout $(0, \pi)$. Hence $\sin x$ is **strictly concave** on $(0, \pi)$.

Step 2: Apply Jensen's Inequality.
For a strictly concave function f and points x_1, x_2, x_3 in its domain, Jensen's inequality states:

$$\frac{f(x_1) + f(x_2) + f(x_3)}{3} \leq f\left(\frac{x_1 + x_2 + x_3}{3}\right).$$

Since $A, B, C \in (0, \pi)$ and $A + B + C = \pi$, we apply this with $x_1 = A$, $x_2 = B$, $x_3 = C$:

$$\frac{\sin A + \sin B + \sin C}{3} \leq \sin\left(\frac{A + B + C}{3}\right) = \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}.$$

Multiplying both sides by 3:

$$\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}. \qquad \square$$

Equality condition. Since $\sin x$ is *strictly* concave, Jensen's inequality becomes an equality if and only if $A = B = C$. Combined with $A + B + C = \pi$, this means $A = B = C = \pi/3$, i.e. the triangle is equilateral. $\square$

Takeaways 3.15

- A function f is concave on an interval whenever $f''(x) \le 0$ throughout that interval.
- Jensen's inequality for a concave f says the average of the function values is at most the function of the average: $\overline{f(x_i)} \le f(\bar{x})$.
- This technique gives tight, elegant bounds on symmetric expressions involving angles of a triangle — a recurring olympiad theme.

Problem 3.16: Roots of Unity and the Cosine Sum

Let $n \ge 2$ be a positive integer and let $\omega = e^{2\pi i/n}$ be a primitive nth root of unity.

(i) By summing the geometric series $1 + \omega + \omega^2 + \cdots + \omega^{n-1}$, prove that

$$\sum_{k=0}^{n-1} \cos\left(\frac{2\pi k}{n}\right) = 0.$$

(ii) Hence evaluate

$$\cos\left(\frac{2\pi}{7}\right) + \cos\left(\frac{4\pi}{7}\right) + \cos\left(\frac{6\pi}{7}\right).$$

Hint 3.16

For part (i), use the fact that $1 + r + r^2 + \cdots + r^{n-1} = \dfrac{r^n - 1}{r - 1}$ for $r \ne 1$. Since $\omega^n = 1$ by definition, the sum equals zero. Take the real part.
For part (ii), split the $n = 7$ sum using the identity from (i), then use $\cos(\pi - x) = -\cos x$.

Solution 3.16

(i) The powers $1, \omega, \omega^2, \ldots, \omega^{n-1}$ form a geometric series with ratio ω and first term 1. Since $\omega \neq 1$ (as $n \geq 2$) and $\omega^n = 1$:

$$\sum_{k=0}^{n-1} \omega^k = \frac{\omega^n - 1}{\omega - 1} = \frac{1-1}{\omega - 1} = 0.$$

By Euler's formula, $\omega^k = e^{2\pi i k/n} = \cos\left(\frac{2\pi k}{n}\right) + i\sin\left(\frac{2\pi k}{n}\right)$. Taking the real part of both sides of the sum:

$$\sum_{k=0}^{n-1} \cos\left(\frac{2\pi k}{n}\right) = \operatorname{Re}(0) = 0. \qquad \square$$

(ii) Apply part (i) with $n = 7$:

$$\sum_{k=0}^{6} \cos\left(\frac{2\pi k}{7}\right) = 0.$$

The $k = 0$ term is $\cos 0 = 1$. The terms for $k = 4, 5, 6$ satisfy

$$\cos\frac{8\pi}{7} = \cos\left(2\pi - \frac{6\pi}{7}\right) = \cos\frac{6\pi}{7}, \quad \cos\frac{10\pi}{7} = \cos\frac{4\pi}{7}, \quad \cos\frac{12\pi}{7} = \cos\frac{2\pi}{7}.$$

(Using $\cos(2\pi - x) = \cos x$.) So:

$$1 + 2\left[\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7}\right] = 0.$$

Therefore:

$$\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7} = -\frac{1}{2}. \qquad \square$$

Takeaways 3.16

- The nth roots of unity $\{e^{2\pi i k/n} : k = 0, \ldots, n-1\}$ always sum to zero; taking real and imaginary parts gives useful trigonometric identities for free.
- The symmetry $\cos(2\pi - x) = \cos x$ pairs up the terms k and $n - k$ in the sum, halving the work for even and odd n.
- This method is the natural Extension 2 companion to the "telescoping cosines" product approach for sums of cosines at equally spaced angles.

4 Conclusion

Trigonometry rewards careful attention to detail as much as formula recall. Many HSC questions become far more tractable once you identify the correct strategy: choosing the right form of $\cos 2\theta$, recognising a products-to-sums pattern, or knowing instinctively when the t-substitution will linearise an equation. As this booklet grows, it will serve as a compact reference for the most important trigonometric patterns students encounter in Extension 1 and Extension 2 examinations and beyond.

Contact Information:

LinkedIn: https://www.linkedin.com/in/nguyenvuhung/

GitHub: https://github.com/vuhung16au/

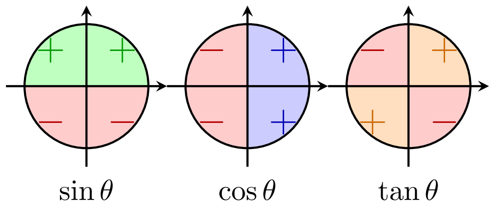
Repository: https://github.com/vuhung16au/math-olympiad-ml/tree/main/HSC-Trigonometry

A Appendices

A.1 The Signs of the Trigonometric Functions

The signs of the trigonometric functions depend only on the signs of x and y. (The radius r is always positive.) The signs of x and y depend in turn only on the quadrant in which the ray lies. The table below summarises the sign of each function in each quadrant; the accompanying diagram shows the coordinate signs on the unit circle.

Function	**1st**	**2nd**	**3rd**	**4th**
x	$+$	$-$	$-$	$+$
y	$+$	$+$	$-$	$-$
r	$+$	$+$	$+$	$+$
$\sin\theta$	$+$	$+$	$-$	$-$
$\cos\theta$	$+$	$-$	$-$	$+$
$\tan\theta$	$+$	$-$	$+$	$-$
$\csc\theta$	$+$	$+$	$-$	$-$
$\sec\theta$	$+$	$-$	$-$	$+$
$\cot\theta$	$+$	$-$	$+$	$-$



$\csc\theta$ has the same sign as $\sin\theta$; $\sec\theta$ has the same sign as $\cos\theta$; $\cot\theta$ has the same sign as $\tan\theta$.

A.2 Triple-Angle Formulae

The double-angle formulae in the Introduction are often enough for routine questions. In harder Extension 1 and Extension 2 problems, it is also useful to recognise the triple-angle formulae:

$$\sin(3\theta) = 3\sin\theta - 4\sin^3\theta,$$

$$\cos(3\theta) = 4\cos^3\theta - 3\cos\theta.$$

These follow by writing $3\theta = 2\theta + \theta$ and applying the compound-angle formulae together with the double-angle formulae. For example,

$$\begin{aligned}\cos(3\theta) &= \cos(2\theta+\theta)\\ &= \cos 2\theta\cos\theta - \sin 2\theta\sin\theta\\ &= (2\cos^2\theta - 1)\cos\theta - (2\sin\theta\cos\theta)\sin\theta\\ &= 4\cos^3\theta - 3\cos\theta.\end{aligned}$$

Similarly,

$$\begin{aligned}\sin(3\theta) &= \sin(2\theta+\theta)\\ &= \sin 2\theta\cos\theta + \cos 2\theta\sin\theta\\ &= 3\sin\theta - 4\sin^3\theta.\end{aligned}$$

Remark A.1 (Connection with Chebyshev polynomials)

The identity

$$\cos(3\theta) = 4\cos^3\theta - 3\cos\theta$$

is the third Chebyshev polynomial in disguise:

$$T_3(x) = 4x^3 - 3x, \qquad T_3(\cos\theta) = \cos(3\theta).$$

More generally, Chebyshev polynomials satisfy $T_n(\cos\theta) = \cos(n\theta)$. This connection is useful when an algebraic polynomial problem can be transformed into a trigonometric equation, as in iterated-polynomial and root-counting questions.

For HSC problem solving, the triple-angle formulae are especially helpful when:

- a cubic expression such as $4x^3 - 3x$ appears and suggests the substitution $x = \cos\theta$;
- an equation involving $\sin 3\theta$ or $\cos 3\theta$ needs to be reduced to powers of $\sin\theta$ or $\cos\theta$;
- a problem asks for exact values or root counts that are difficult to see from algebra alone.

A.3 Derivatives of Inverse Trigonometric Functions

The standard derivative formulae for the inverse trigonometric functions are:

$$\frac{d}{dx}\big(\arcsin x\big) = \frac{1}{\sqrt{1-x^2}}, \qquad -1 < x < 1,$$

$$\frac{d}{dx}\big(\arccos x\big) = -\frac{1}{\sqrt{1-x^2}}, \qquad -1 < x < 1,$$

$$\frac{d}{dx}\big(\arctan x\big) = \frac{1}{1+x^2}, \qquad x \in \mathbb{R}.$$

These are useful in Extension 2 calculus questions involving inverse trigonometric functions, especially when differentiating composite functions. For example,

$$\frac{d}{dx}\arcsin(2x-1) = \frac{2}{\sqrt{1-(2x-1)^2}},$$

where the expression is defined.

A.4 Taylor Series for Sine and Cosine

The following Maclaurin series are out of the HSC syllabus, but they are very useful in university mathematics, physics, engineering, and numerical methods:

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots,$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots.$$

These expansions show how trigonometric functions can be approximated by polynomials near $x = 0$. For instance, when x is small,

$$\sin x \approx x, \qquad \cos x \approx 1 - \frac{x^2}{2}.$$

A.5 Introduction to the Laplace Transform

Remark A.2

The Laplace transform is beyond the NSW HSC Extension 1 and Extension 2 syllabuses. It is included here as enrichment for students considering university mathematics, engineering, or physics.

The **Laplace transform** converts a function of time t into a function of a parameter s, turning differential equations into algebraic ones. For a function $f(t)$ defined for $t \geq 0$:

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t)\, dt,$$

provided the improper integral converges (which requires s to be sufficiently large).

Example 1: Laplace transform of $\cos(at)$. For $a > 0$ and $s > 0$:

$$\mathcal{L}\{\cos(at)\} = \int_0^\infty e^{-st} \cos(at)\, dt = \frac{s}{s^2 + a^2}.$$

The derivation uses integration by parts twice (the "looping" integral), or Euler's formula. See the worked problem in Part 1 Advanced for the full derivation.

Example 2: Laplace transform of $\sin(at)$.

$$\mathcal{L}\{\sin(at)\} = \int_0^\infty e^{-st} \sin(at)\, dt = \frac{a}{s^2 + a^2}.$$

This follows simultaneously from the complex-exponential method: the imaginary part of $\mathcal{L}\{e^{iat}\} = \dfrac{1}{s - ia}$ gives the sine transform for free.

Why it matters. The Laplace transform turns the differential equation $y'' + \omega^2 y = 0$ (simple harmonic motion) into the algebraic equation $(s^2 + \omega^2)Y(s) =$ initial conditions. Inverting using the table above recovers $y(t) = A\cos(\omega t) + B\sin(\omega t)$—exactly the sinusoidal solutions that arise throughout HSC physics.

